



ALIRO Specification Test Plan

Version 1.0

Document:	26-42803-001_Aliro_1.0_specification_test_plan
	February 9, 2026
Sponsored by:	Connectivity Standard Alliance
Accepted by:	This document has been accepted for release by the Connectivity Standards Alliance Board of Directors
Abstract	This test plan describes the tests to be performed for supported Aliro 1.0 features.

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Revision history

Revision	Date	Details	Editor
1.0	Jan 2026	release 1.0	

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1 Introduction

1.1 Scope

This test plan covers the tests for Aliro 1.0.

1.2 Purpose

Describes tests to be performed for supported Aliro 1.0 features.

1.3 Provisional Status Notification

This section exists to inform of the results implications of the validation process on technical specification. As per Connectivity Standards Alliance Policies & Procedures, the following items are marked as provisional based on the results of the Aliro Standard Validation Event resolution.

- [Optional] User Device test as in 7.13 Select Response with User Device Descriptor Tag
- [Optional] User Device test as in 7.40 BLE+UWB Flow with User Device Descriptor Tag
- [Optional] User Device test as in 7.50 BLE-Only Flow with Expedited Standard Phase
- [Optional] User Device test as in 7.51 BLE-Only Flow with User Device Descriptor Tag
- [Optional] User Device test as in 7.52 BLE-Only Flow with Failed L2CAP

2 References

- [1] ALIRO Specification Version 0.9.0

3 Definitions

3.1 Acronyms

The acronyms are defined in Aliro 1.0 [1].

3.2 Glossary

Access Credential	A set of information that contains all data necessary to perform the access transaction, this includes the Access Credential key pair and an optional Access Document.
Access Data Element	Standardized structure to define access information.
Access Document	Issued by a Credential Issuer. Contains Access Data Elements and the Access Credential public key.
Access Manager	A manager used to determine whether access should be granted and actuate locking mechanisms. An Access Manager may be embedded in the Reader.
Credential Issuer	Issuer of the Access Credential.
Reader	A reader device to read an Access Credential from a User Device and optionally send Access Credential information to an Access Manager.
Reader System Issuer	Issuer of the Reader.
Revocation Data Element	Standardized structure to define revocation information
Revocation Document	Issued by the Credential Issuer. Contains revocation data elements.
User Device	A portable device containing one or more access credentials E.g., card, fob, tag, key, mobile phone, smartwatch etc.

3.3 Conformance Levels

The key word meaning is defined in Aliro 1.0 [1].

4 Test Setup

4.1 Architecture

Figure 4-1 provides a detailed illustration of the Aliro Test Harness. This system is designed to execute test scripts that align with the test plan. Within this setup, the Aliro Actuator represents the implementation of the Aliro technical specifications.

The Test Harness is built around the Raspberry Pi development platform. It interfaces with various external platforms: UWB, BLE, and NFC. Specifically, it employs the NXP Murata SR150 UWB platform, which is connected via a USB interface using UCI-PnP. For NFC operations, the Test Harness utilizes the NXP PN7160 Reader/Card Emulation NFC platform, connected through an SPI or I2C interface, depending on the model of the NFC platform. The Bluetooth LE in the Murata board is used and not the Raspberry Pi's built-in Bluetooth module for the performance purposes.

The Test Harness communicates wirelessly with the Device Under Test (DUT) using the Aliro Protocol.

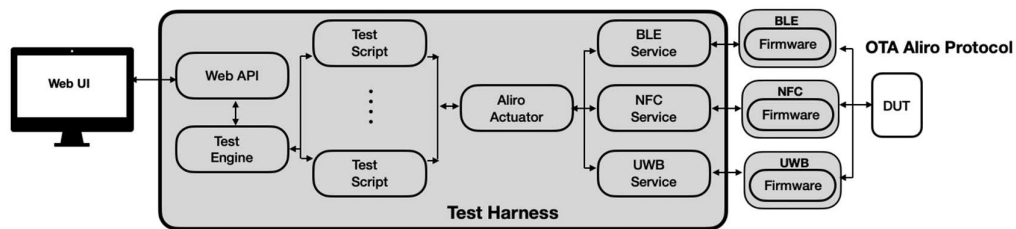


Figure 4-1 Aliro Test Harness architecture

4.1.1 PICS

In this section the PICS parameter requirements for Reader and User Device are enumerated. ‘M’ implies mandatory feature, ‘O’ implies optional feature, ‘C’ implies conditional feature, and ‘NA’ implies feature Not Applicable for the purposes of tests. PICS parameter with a corresponding empty cell implies no test is defined.

Table 4-1 Expedited-Standard phase PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Expedited-Standard Phase	M	M	NFC_RDR_STANDARD_NO_CERT, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, , NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING,	NFC_UD_STANDARD_NO_CERT, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, , NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING,

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
			NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING, NFC_RDR_NEG_SEL_RSP_NO_COMMON_EXPEDITED_PROTOCOL_VERSION, NFC_RDR_NEG_AUTH0_EXTRA_TAG, NFC_RDR_NEG_AUTH0_WRONG_VALUE, NFC_RDR_NEG_AUTH1_WRONG_UD_SIGNATURE, NFC_RDR_NEG_AUTH1_EXTRA_TAG, NFC_RDR_NEG_AUTH1_WRONG_VALUES	NFC_UD_AUTH0_RESPONSE_CHAINING, NFC_UD_NEG_AUTH0_UNKNOWN_READER_ID, NFC_UD_NEG_AUTH0_UNSUPPORTED_PROTOCOL_VERSION, NFC_UD_NEG_AUTH0_EXTRA_TAG, NFC_UD_NEG_AUTH0_WRONG_VALUE, NFC_UD_NEG_AUTH0_WRONG_P1P2, NFC_UD_NEG_AUTH0_CHAINING_NOT_COMPLETED, NFC_UD_NEG_AUTH1_WRONG_READER_SIGNATURE, NFC_UD_NEG_AUTH1_EXTRA_TAG, NFC_UD_NEG_AUTH1_WRONG_P1P2, NFC_UD_NEG_AUTH1_WRONG_VALUES
Reader signature generation and validation using reader_PubK	M	M	NFC_RDR_STANDARD_NO_CERT	NFC_UD_STANDARD_NO_CERT, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert)	M	M	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
Device signature generation and validation	M	M	NFC_RDR_STANDARD_NO_CERT, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING,	NFC_UD_STANDARD_NO_CERT, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING,

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
			NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
Verification of reader_Cert with the CA public key	NA	M	NA	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING, NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_SIGNATURE, NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_FORMAT
Verification of reader_Cert with the CA public key – reader_Cert expiration time validation	NA	O	NA	
Lookup of the reader key through reader_group_identifier	NA	M	NA	NFC_UD_STANDARD_NO_CERT, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING
Lookup of reader CA public key through reader_group	NA	M	NA	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
up_identifier				
Presentation and validation of reader_Cert	M	M	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
Presentation and validation of reader_Cert in AUTH1 command	C	M	NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
Presentation and validation of reader_Cert in LOAD_CERT command	C	M	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING
AUTH1 command command_parameter	M	M	NFC_RDR_STANDARD_NO_CERT, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_NO_CERT, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
AUTH1 command	C	M	NFC_RDR_STANDARD_NO_CERT,	NFC_UD_STANDARD_NO_CERT,

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
command_parameter – key slot			NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
AUTH1 command_parameter – Access Credential Public Key	C	M	NFC_RDR_STANDARD_NO_CERT, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_NO_CERT, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING
Subset of mailbox in AUTH1 command response	NA	M	NA	

Table 4-2 Expedited-Fast phase PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Expedited-Fast Phase	O	O	NFC_RDR_FAST	NFC_UD_FAST, NFC_UD_NEG_AUTH0_DIFFERENT_CRYPTOGAM_CONSECUTIVE_FAST
Cryptogram generation and validation	M	M	NFC_RDR_FAST	NFC_UD_FAST, NFC_UD_NEG_AUTH0_DIFFERENT_CRYPTOGAM_CONSECUTIVE_FAST

Table 4-3 Expedited Phase PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Command chaining	M	M	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING, NFC_UD_AUTH0_RESPONSE_CHAINING, NFC_UD_NEG_AUTH1_CHAINING_NOT_COMPLETED,
Extended length	O	O	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING, NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING, NFC_UD_EXCHANGE_WITH_EXTENDED_LENGTH
User authentication policy enforcement	NA	M	NA	NFC_UD_STANDARD_NO_CERT
Support for credential_signed_timestamp in AUTH0/AUTH1	NA	M		
Support for revocation_signed_timestamp in AUTH0/AUTH1	NA	M		
Allow at least 16 reader_group_identifier per Access Credential	NA	M	NA	NFC_UD_STANDARD_SIXTEEN_GROUPIDENTIFIER_ONE_AC

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
User Device has a method to use all Access Credentials bound to the reader identifier	NA	M	NA	
EXCHANGE command	M	M		NFC_UD_EXCHANGE_WITH_CHAINING, NFC_UD_NEG_EXCHANGE_WITH_EXTRA_TAG, NFC_UD_NEG_EXCHANGE_WITH_WRONG_LENGTH
EXCHANGE command – notify credential issuer in EXCHANGE	O	O		
EXCHANGE command – notify bound application in EXCHANGE	O	O		
EXCHANGE command – Update document in EXCHANGE	O	O		
EXCHANGE command – Update document in EXCHANGE – providing and processing update_doc contents	NA	NA	NA	NA
Mailbox	O	M	NFC_RDR_EXCHANGE_MAILBOX	NFC_UD_EXCHANGE_SET_REQUEST,

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
				NFC_UD_NEG_EXCHANGE_MAILBOX_OUT_OF_BOUNDS
Mailbox Read/Write by credential issuer	NA	NA	NA	NA
Read from Mailbox	O	M		NFC_UD_EXCHANGE_READ_REQUEST
Write to Mailbox	O	M		NFC_UD_EXCHANGE_WRITE_REQUEST
Reader Descriptor tag	O	NA	NFC_RDR_EXCHANGE_RDR_DESCRIPTOR_TAG, NFC_RDR_EXCHANGE_RDR_DESCRIPTOR_TAG, NFC_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG, BLEUWB_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG	NA
User Device Descriptor Tag	NA	O	NA	NFC_UD_SELECT_RESPONSE_UD_DESCRIPTOR_TAG, BLEUWB_UD_UD_DESCRIPTOR_TAG, BLERKE_UD_UD_DESCRIPTOR_TAG

Table 4-4 Step-Up Phase PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Step-Up phase	O	M	NFC_RDR_STEPUP_AD_KEY_ID, NFC_RDR_STEPUP_AD_ISSUER_CERT, NFC_RDR_STEPUP_AD_ISSUER_CERT_KEY_ID, NFC_RDR_STEPUP_AD_ACCESS_RULE, NFC_RDR_STEPUP_AD_ACCESS_RULE_SCHEDULES, NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION, NFC_RDR_STEPUP_AD_UNKNOWN_NON_CRITICAL_ACCESS_EXTENSION, NFC_RDR_NEG_STEPUP_AD_NO_ISSUER_CERT_NO_KEY_ID,	NFC_UD_STEPUP_AD, NFC_UD_STEPUP_RD

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
			NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_INVALID_SIGNATURE, NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_EXPIRED, NFC_RDR_NEG_STEPUP_AD_INVALID_SIGNATURE_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_INVALID_HASH_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_EXPIRED_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_EARLY_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_ISSUER_CERTIFICATE_TIME_MISMATCH, NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION, NFC_RDR_NEG_STEPUP_AD_TIME_VERIFICATION_REQUIRED, NFC_RDR_NEG_STEPUP_AD_NO_DATA_ELEMENTS, NFC_RDR_NEG_STEPUP_AD_ISSUER_DOCTYPE_MISMATCH, NFC_RDR_NEG_STEPUP_AD_DEVICE_KEY_INFO_MISMATCH, NFC_RDR_NEG_STEPUP_AD_INVALID_ACCESS_DATA_ELEMENT_VERSION, NFC_RDR_NEG_STEPUP_AD_NO_ACCESS_RULE_FOR_READER_ACTION, NFC_RDR_NEG_STEPUP_AD_NO_VALID_SCHEDULE_ALLOW_SCHEDULEID, NFC_RDR_NEG_STEPUP_AD_VALID_SCHEDULE_DENY_SCHEDULEID, NFC_RDR_NEG_STEPUP_AD_SCHEDULE_TIME_VERIFY_REQUIRED, NFC_RDR_NEG_STEPUP_AD_SCHEDULE_IN_ACCESS_RULE_AND_READER, NFC_RDR_NEG_STEPUP_AD_UNKNOWN_READER_RULE, NFC_RDR_NEG_STEPUP_AD_UNKNOWN_CRITICAL_ACCESS_EXTENSION, NFC_RDR_STEPUP_RD, NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION	
Step-Up phase – Access document storage	O	M	NFC_RDR_STEPUP_AD_KEY_ID	NFC_UD_STEPUP_AD

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
and retrieval				
Step-Up phase – Revocation document storage and retrieval	O	M	NFC_RDR_STEPUP_RD, NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION	NFC_UD_STEPUP_RD

Table 4-5 Access Document processing PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Access document processing	O	NA	NFC_RDR_STEPUP_AD_ISSUER_CERT, NFC_RDR_STEPUP_AD_ISSUER_CERT_KEY_ID, NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION, NFC_RDR_STEPUP_AD_UNKNOWN_NON_CRITICAL_ACCESS_EXTENSION, NFC_RDR_NEG_STEPUP_AD_NO_ISSUER_CERT_NO_KEY_ID, NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_INVALID_SIGNATURE, NFC_RDR_NEG_STEPUP_AD_INVALID_SIGNATURE_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_INVALID_HASH_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_ISSUER_CERTIFICATE_TIME_MISMATCH, NFC_RDR_NEG_STEPUP_AD_NO_DATA_ELEMENTS, NFC_RDR_NEG_STEPUP_AD_ISSUER_DOCTYPE_MISMATCH, NFC_RDR_NEG_STEPUP_AD_DEVICE_KEY_INFO_MISMATCH, NFC_RDR_NEG_STEPUP_AD_INVALID_ACCESS_DATA_ELEMENT_VERSION, NFC_RDR_NEG_STEPUP_AD_NO_ACCESS_RULE_FOR_READER_ACTION, NFC_RDR_NEG_STEPUP_AD_UNKNOWN_READER_RULE, NFC_RDR_NEG_STEPUP_AD_UNKNOWN_CRITICAL_ACCESS_EXTENSION	NA

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Access document verification	M	NA	NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_EXPIRED, NFC_RDR_NEG_STEPUP_AD_EXPIRED_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_EARLY_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION, NFC_RDR_NEG_STEPUP_AD_TIME_VERIFICATION_REQUIRED	NA
Access document verification – Validity iteration	M	NA	NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION	NA
Access document verification – Validity time-based elements	O	NA	NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_EXPIRED, NFC_RDR_NEG_STEPUP_AD_EXPIRED_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_EARLY_ISSUER_AUTH, NFC_RDR_NEG_STEPUP_AD_TIME_VERIFICATION_REQUIRED , NFC_RDR_NEG_STEPUP_AD_SCHEDULE_TIME_VERIFY_REQUIRED	NA
Access data element verification	M	NA	NFC_RDR_STEPUP_AD_ACCESS_RULE, NFC_RDR_STEPUP_AD_ACCESS_RULE_SCHEDULES	NA
Access data element verification – Access Rules	M	NA	NFC_RDR_STEPUP_AD_ACCESS_RULE	NA
Access data element verification –	O	NA	NFC_RDR_STEPUP_AD_ACCESS_RULE_SCHEDULES, NFC_RDR_NEG_STEPUP_AD_NO_VALID_SCHEDULE_ALLOW_SCHEDULEID, NFC_RDR_NEG_STEPUP_AD_VALID_SCHEDULE_DENY_SCHEDULEID,	NA

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Schedules			NFC_RDR_NEG_STEPUP_AD_SCHEDULE_IN_ACCESS_RULE_AND_READER	
Access data element verification – Access extension criticality	M	NA		NA
Access data element verification – Access extension content	NA	NA	NA	NA
Access data element verification – Non access extension	NA	NA	NA	NA
Access data element verification – Reader rules	O	NA	NFC_RDR_NEG_STEPUP_AD_UNKNOWN_READER_RULE	NA
Access data element verification - ID	NA	NA	NA	NA

Table 4-6 Revocation document processing PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Revocation document processing	O	NA	NFC_RDR_STEPUP_RD, NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION	NA
Revocation document verification	M	NA	NFC_RDR_STEPUP_RD, NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION	NA
Revocation element verification	M	NA	NFC_RDR_STEPUP_RD, NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION	NA

Table 4-7 NFC interface PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
NFC Interface	M	M	Table 4-1	Table 4-1
NFC – Step-Up AID SELECT	M	O	NFC_RDR_STEPUP_AD_KEY_ID	
Vendor-specific extensions in SELECT	NA	NA	NA	NA

Table 4-8 BLE interface PICS parameters

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
BLE Interface	O	O	Table 4-9, Table 4-10	Table 4-9, Table 4-10

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
BLE – send sensor triggered bit	O	O		
Dynamic advertisement tag	M	O	BLEUWB_RDR_EXPEDITED_STANDARD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE, BLEUWB_RDR_ADVERTISEMENT_FORMAT	
Pass Through	O	O		
Unsolicited Reader status reporting	M	NA	BLEUWB_RDR_EXPEDITED_STANDARD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE, BLERKE_RDR_UNSECURE, BLERKE_RDR_SECURE	NA

Table 4-9 BLE + UWB interface PICS parameters for Bluetooth LE + UWB Aliro Flow

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
Bluetooth LE + UWB Flow	O	O	BLEUWB_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG, BLEUWB_RDR_EXPEDITED_STANDARD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE, BLEUWB_RDR_RANGING_SUSPEND, BLEUWB_RDR_RANGING_RESUME, BLEUWB_RDR_NEG_FAILED_L2CAP, BLEUWB_RDR_NEG_FAILED_SPSM_L2CAP, BLEUWB_RDR_NEG_TIMEOUT_BEFORE_AUTH0, BLEUWB_RDR_TIMEOUT_EXTENSION, BLEUWB_RDR_NEG_M2_MISMATCH_PARAMETER,	BLEUWB_UD_EXPEDITED_STANDARD_PHASE, BLEUWB_UD_EXPEDITED_FAST_PHASE, BLEUWB_UD_STEPUP_PHASE, BLEUWB_UD_RANGING_SUSPEND, BLEUWB_UD_RANGING_RESUME, BLEUWB_UD_DESCRIPTOR_TAG, BLEUWB_UD_NEG_WRONG_ADV, BLEUWB_UD_NEG_FAILED_L2CAP, BLEUWB_UD_NEG_TIMEOUT_BEFORE_AUTH0, BLEUWB_UD_TIMEOUT_EXTENSION, BLEUWB_UD_NEG_URSK_NOT_FOUND,

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
			BLEUWB_RDR_NEG_M4_MISMATCH_PARAMETER, BLEUWB_RDR_NEG_SUSPEND_MISMATCH_PARAMETER, BLEUWB_RDR_ADVERTISEMENT_FORMAT	BLEUWB_UD_NEG_M1_MISMATCH_PARAMETER, BLEUWB_UD_NEG_M3_MISMATCH_PARAMETER
UWB ranging	M	M	BLEUWB_RDR_EXPEDITED_STANDARDD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE	BLEUWB_UD_EXPEDITED_STANDARDD_PHASE, BLEUWB_UD_EXPEDITED_FAST_PHASE, BLEUWB_UD_STEPUP_PHASE
UWB ranging suspend	M	M	BLEUWB_RDR_RANGING_SUSPEND, BLEUWB_RDR_NEG_SUSPEND_MISMATCH_PARAMETER	BLEUWB_UD_RANGING_SUSPEND, BLEUWB_UD_NEG_SUSPEND_MISMATCH_PARAMETER
UWB ranging resume	M	M	BLEUWB_RDR_RANGING_RESUME	BLEUWB_UD_RANGING_RESUME, BLEUWB_UD_NEG_RESUME_MISMATCH_PARAMETER
One ranging round	M	M	BLEUWB_RDR_EXPEDITED_STANDARDD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE	BLEUWB_UD_EXPEDITED_STANDARDD_PHASE, BLEUWB_UD_EXPEDITED_FAST_PHASE, BLEUWB_UD_STEPUP_PHASE
Two ranging rounds	O	M	BLEUWB_RDR_EXPEDITED_STANDARDD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE	BLEUWB_UD_EXPEDITED_STANDARDD_PHASE, BLEUWB_UD_EXPEDITED_FAST_PHASE, BLEUWB_UD_STEPUP_PHASE
BLE UWB Time synchronization	O	M	BLEUWB_RDR_EXPEDITED_STANDARDD_PHASE, BLEUWB_RDR_EXPEDITED_FAST_PHASE, BLEUWB_RDR_STEPUP_PHASE	BLEUWB_UD_EXPEDITED_STANDARDD_PHASE, BLEUWB_UD_EXPEDITED_FAST_PHASE, BLEUWB_UD_STEPUP_PHASE

Table 4-10 BLE interface PICS parameters for BLE-Only Flow

PICS Parameter	Reader	User Device	Reader Test Identifier	User Device Test Identifier
BLE-Only Flow	O	O	BLERKE_RDR_UNSECURE, BLERKE_RDR_SECURE, BLERKE_RDR_NEG_FAST, BLERKE_RDR_NEG_FAILED_L2CAP, BLERKE_RDR_NEG_FAILED_SPSM_L2CAP, BLERKE_RDR_STEPUP_PHASE	BLERKE_UD_EXPEDITED_STANDAR D_PHASE, BLERKE_UD_UD_DESCRIPTOR_TAG, BLERKE_UD_NEG_FAILED_L2CAP
Explicit Reader selection	NA	M	NA	BLERKE_UD_EXPEDITED_STANDAR D_PHASE

5 User Device Under Test Routines

This section describes routines used in User Device Under Test tests.

5.1 SELECT Routine

Table 5-1 SELECT routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send SELECT command		
2		send SELECT response	Verify the following: - order of TLVs matches the technical specification - All mandatory TLVs in technical specification are present 0100h is present in the expedited_phase_supported_protocol_versions. - size of SELECT response is less than 256B - AID = A000000909ACCE5501, if Expedited Phase - AID = A000000909ACCE5502, if Step-Up Phase - Type = 0000h - Unknown TLVs are ignored, if present If all criteria are met, then CONTINUE else FAIL.

5.2 AUTH0 Routine

Table 5-2 AUTH0 routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send AUTH0 command - authentication_policy = select randomly between 01h, 02h, 03h - expedited_phase_protocol_version = 0100h		Verify the following: user authentication is performed by User Device, if authentication_policy = 03h. If all criteria are met, then CONTINUE else FAIL.
2		send AUTH0 response	Verify the following: - order of TLVs matches technical specification - All mandatory TLVs in technical specification are present - cryptogram is not present, if command_parameters = 0h - cryptogram is present, if command_parameters = 1h - auth0_response_vendor_extension, if present is less than 127B in size

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			6. SW = 9000h 9. Success in establishing secure channel 10. Unknown TLVs are ignored, if present If all criteria are met, then CONTINUE else FAIL.

5.3 AUTH1 with SW Equal to 9000h Routine

Table 5-3 AUTH1 with SW = 9000h routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
5	Send AUTH1 command 1. command_parameters = randomly selected between 00h, 01h		
6		send AUTH1 response	Verify the following: 1. order of TLVs matches technical specification. 2. All mandatory TLVs in technical specification are present. 3. key_slot is present, if command_parameters = 01h. 4. Access Credential long term public key is present, if command_parameters = 00h. 5. SW = 9000h. 6. User Device signature verification passes. 7. Unknown TLVs are ignored, if present If all criteria are met, then CONTINUE else FAIL.

5.4 AUTH1 with SW Not Equal to 9000h Routine

Table 5-4 AUTH1 with SW != 9000h routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send AUTH1 command 1. command_parameters = randomly selected between 00h, 01h		
2		send AUTH1 response	Verify the following: 1. SW != 9000h.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.

5.5 EXCHANGE Indicating Transaction Success Routine

Table 5-5 EXCHANGE indicating transaction success routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send EXCHANGE command with tag 0x97 (Reader Status).		Verify the following: 1. 0x97h first byte is 0x01h. If all criteria are met, then CONTINUE else FAIL.
2		send EXCHANGE response.	Verify the following: 1. response payload = 0x0002 0x00 0x00. 2. SW = 9000h. If all criteria are met, then PASS else FAIL.

5.6 EXCHANGE Indicating Transaction Failure Routine

Table 5-6 EXCHANGE indicating transaction failure routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send EXCHANGE command with tag 0x97 (Reader Status).		Verify the following: 0x97h first byte is 0x00h. If all criteria are met, then CONTINUE else FAIL.
2		send EXCHANGE response.	Verify the following: 1. response payload = 0x0002 0x00 0x00. 2. SW = 9000h. If all criteria are met, then PASS else FAIL.

5.7 CONTROL FLOW Indicating Transaction Failure Routine

Table 5-7 CONTROL FLOW indicating transaction failure routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send CONTROL FLOW command.		Verify the following: CONTROL FLOW command data field length does not exceed 255 bytes. CONTROL FLOW command is formatted according to the specification. If all criteria are met, then CONTINUE else FAIL.
2		send CONTROL FLOW response.	Verify the following: 1. response data field is empty. 2. SW = 9000h. If all criteria are met, then PASS else FAIL.

5.8 BLE+UWB Aliro Access Protocol Routine

Table 5-8 BLE+UWB Aliro Access Protocol routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement		
2		Establish L2CAP connection	
3		Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501	Verify the following: Format of Initiate Access Protocol Message ID matches specification. If all criteria are met, then CONTINUE else FAIL.
4	Execute AUTH0 routine. Set command_parameters = 0h and authentication_policy = 01h (User Device)		If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH1 routine with SW = 9000h. Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
6	Send EXCHANGE command		Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
7		Send EXCHANGE response	
8	Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID		Verify the following: Ensure reader status is secured. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.

5.9 BLE+UWB Ranging Session Setup Routine

Table 5-9 BLE+UWB ranging session setup routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Ranging Session Setup M1 Message ID		
2		send Ranging Session Setup M2 Message ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
3	Send Ranging Session Setup M3 Message ID		
4		send Ranging Session Setup M4 Message ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.

5.10 BLE-Only Aliro Access Protocol Routine

Table 5-10 BLE-only Aliro Access Protocol routine

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement.		
2		Establish L2CAP connection.	
3		User register intent to perform RKE action.	Verify the following: User selects the Reader and indicates the action.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			Note: how the user selects the Reader, and the associated action is implementation choice. If all criteria are met, then CONTINUE else FAIL.
4		Send Initiate Access Protocol RKE Message ID carrying AID = A000000909ACCE5501	Verify the following: Format of Message ID matches the technical specification. If all criteria are met, then CONTINUE else FAIL.
5	Send AUTH0 command command_parameters = 0h authentication_policy = 01h (User Device) or 03h (Force User Auth)		Verify the following: User Auth is performed, if authentication_policy = 03h. If all criteria are met, then CONTINUE else FAIL.
6		Send AUTH0 response	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
7	Send AUTH1 command and reader_cert is not present.		
		send AUTH1 response	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
8	Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID		Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.

6 Reader Under Test Routines

This section describes routines used in Reader Under Test tests.

6.1 SELECT Routine

Table 6-1 SELECT routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send SELECT command	Verify the following: 1. AID = A000000909ACCE5501, if Expedited Phase 2. AID = A000000909ACCE5502, if Step-Up Phase If all criteria are met, then CONTINUE else FAIL.
2	send SELECT response		

6.2 AUTH0 Routine

Table 6-2 AUTH0 routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send AUTH0 command	Verify the following: 1. order of TLVs in AUTH0 command matches specification. 2. All mandatory TLVs in AUTH0 command are present 3. expedited_phase_protocol_version = 0100h 4. Unknown TLVs are ignored, if present If all criteria are met, then CONTINUE else FAIL.
2	Send AUTH0 response		

6.3 AUTH1 Routine

Table 6-3 AUTH1 routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send AUTH1 command	Verify the following: 1. order of TLVs in AUTH1 command matches specification. 2. All mandatory TLVs in AUTH1 command are present 3. Unknown TLVs are ignored, if present If all criteria are met, then CONTINUE else FAIL.
2	Send AUTH1 response		

6.4 EXCHANGE Indicating Transaction Success Routine

Table 6-4 EXCHANGE indicating transaction success routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		send EXCHANGE command with tag 0x97 (Reader Status)	Verify the following: - order of TLVs in EXCHANGE command matches specification. - All mandatory TLVs in EXCHANGE command are present 0x97h first byte is 0x01h. - Unknown TLVs are ignored, if present If all criteria are met, then CONTINUE else FAIL.
2	Send EXCHANGE response		

6.5 EXCHANGE Indicating Transaction Failure Routine

Table 6-5 EXCHANGE indicating transaction failure routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		send EXCHANGE command with tag 0x97 (Reader Status)	Verify the following: 0x97h first byte is 0x00h. If all criteria are met, then CONTINUE else FAIL.
2	Send EXCHANGE response		

6.6 CONTROL FLOW Indicating Transaction Failure Routine

Table 6-6 CONTROL FLOW indicating transaction failure routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send CONTROL FLOW command.	Verify the following: - command data field length does not exceed 255 bytes. - Format of CONTROL FLOW command matches the specification. If all criteria are met, then CONTINUE else FAIL.
2	send CONTROL FLOW response.		Verify the following: - response data field is empty. - SW = 9000h. If all criteria are met, then PASS else FAIL.

6.7 BLE+UWB Aliro Access Protocol Routine

Table 6-7 BLE+UWB Aliro Access Protocol routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is set to 1. Advertisement format matches the technical specification. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection		
3	Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501		
4	Execute AUTH0 routine		If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH1 routine		If all criteria are met, then CONTINUE else FAIL.
8		Send EXCHANGE command	Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.
9	Send EXCHANGE response		
10		Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID	Verify the following: reader status is secured. If all criteria are met, then CONTINUE else FAIL.

6.8 BLE+UWB Ranging Session Setup Routine

Table 6-8 BLE+UWB ranging session setup routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
2		Send Ranging Session Setup M1 Message ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
3	send Ranging Session Setup M2 Message ID		

Steps	TH (User Device)	DUT (Reader)	Verification at TH
4		send Ranging Session Setup M3 Message ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
5	send Ranging Session Setup M4 Message ID		

6.9 BLE-Only Aliro Access Protocol Routine

Table 6-9 BLE-only Aliro Access Protocol routine

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE-Only Aliro Flow Supported Bit is set to 1. Advertisement format matches the technical specification. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection		
3	Send Initiate Access Protocol RKE Message ID carrying AID = A000000909ACCE5501		
4		Send AUTH0 command	Verify the following: command_parameters = 0h. If all criteria are met, then CONTINUE else FAIL.
5	send AUTH0 response		
6		Send AUTH1 command	
7	Send AUTH1 response		
8		[Optional] Send EXCHANGE command	
9	Send EXCHANGE response		Sent, if an EXCHANGE command is received.
10		Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID	Verify the following: Format matches technical specification. If all criteria are met, then CONTINUE else FAIL.

7 User Device Under Test Conformance Tests

7.1 Expedited Standard Phase without Reader Certificate

Table 7-1 NFC_UD_STANDARD_NO_CERT test identifiers

Parameter	Value
Test ID	NFC_UD_STANDARD_NO_CERT
PICS	Expedited-Standard Phase AND User Authentication Policy Enforcement AND Reader signature generation and validation using reader_PubK AND Device signature generation and validation AND Lookup of the reader key through reader_group_identifier AND AUTH1 command parameter
Applicability	M for User Device
Interface	NFC

Table 7-2 NFC_UD_STANDARD_NO_CERT test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier
TH (Reader)	Access Credential long term public key

Table 7-3 NFC_UD_STANDARD_NO_CERT test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h. auth0_command_vendor_extension is present in AUTH0 command.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.2 Expedited Standard Phase with Reader Certificate in LOAD_CERT with APDU Chaining

Table 7-4 NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test identifiers

Parameter	Value
Test ID	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING
PICS	Expedited-Standard Phase AND Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Presentation and validation of the reader_Cert in LOAD_CERT command Device signature generation and validation AND Verification of the reader_Cert with the CA Public Key AND Lookup of the reader CA Public Key through reader_group_identifier AND AUTH1 command parameter AND Command chaining
Applicability	M for User Device
Interface	NFC

Table 7-5 NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions

Provision onto	Remarks
DUT (User Device)	Reader System Issuer CA public key, reader_group_identifier
TH (Reader)	Access Credential long term public key, reader_Cert

Table 7-6 NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Send LOAD_CERT command with fragmented reader_cert with chaining.		
4		Send LOAD_CERT response.	Verify the following: 1. SW =9000h. 2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.3 Expedited Standard Phase with Reader Cert in LOAD_CERT without APDU Chaining

Table 7-7 NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING test identifiers

Parameter	Value
Test ID	NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING
PICS	Expedited-Standard Phase AND Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Presentation and validation of the reader_Cert in LOAD_CERT command Device signature generation and validation AND Verification of the reader_Cert with the CA Public Key AND Lookup of the reader CA Public Key through reader_group_identifier AND AUTH1 command parameter AND Extended length
Applicability	M for User Device, if it supports Extended length APDUs
Interface	NFC

NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING test pre-conditions are identical to NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 7-5.

Table 7-8 NFC_UD_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING test steps

Step#	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h		If all criteria are met, then CONTINUE else FAIL.
3	Send LOAD_CERT command with reader_cert and no APDU chaining.		
4		Send LOAD_CERT response.	Verify the following: 1. SW =9000h. 2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.4 Expedited Standard Phase with Reader Cert in AUTH1 with APDU Chaining

Table 7-9 NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING test identifiers

Parameter	Value
Test ID	NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING
PICS	Expedited-Standard Phase AND Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Presentation and validation of the reader_Cert in AUTH1 command Device signature generation and validation AND Verification of the reader_Cert with the CA Public Key AND Lookup of the reader CA Public Key through reader_group_identifier AND AUTH1 command parameter AND Command chaining
Applicability	M for User Device
Interface	NFC

NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING test pre-conditions are identical to NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 7-5.

Table 7-10 NFC_UD_STANDARD_CERT_IN_AUTH1_WITH_CHAINING test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is present and fragmented with chaining over multiple APDUs.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.5 Expedited Standard Phase with Reader Cert in AUTH1 without APDU Chaining

Table 7-11 NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING test identifiers

Parameter	Value
Test ID	NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING

PICS	Expedited-Standard Phase AND Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Presentation and validation of the reader_Cert in AUTH1 command Device signature generation and validation AND Verification of the reader_Cert with the CA Public Key AND Lookup of the reader CA Public Key through reader_group_identifier AND AUTH1 command parameter AND Extended length
Applicability	M for User Device, if it supports Extended length APDU
Interface	NFC

NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING test pre-conditions are identical to NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 7-5.

Table 7-12 NFC_UD_STANDARD_CERT_IN_AUTH1_NO_CHAINING test steps

Step#	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is present and without chaining over multiple APDUs.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.6 Expedited Fast Phase

Table 7-13 NFC_UD_FAST test identifiers

Parameter	Value
Test ID	NFC_UD_FAST
PICS	Expedited-fast
Applicability	M for User Device that supports Expedited Fast Phase
Interface	NFC

NFC_UD_FAST test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-14 NFC_UD_FAST test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		Note down reader_group_sub_identifier. If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5)..		If all criteria are met, then CONTINUE else FAIL.
5	Wait for at least 3 seconds.		
6	Execute SELECT routine. Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
7	Execute AUTH0 routine. Set command_parameters = 1h. Use reader_group_identifier and reader_group_sub_identifier from step 2.		If all criteria are met, then CONTINUE else FAIL.
8	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.7 Expedited Standard Phase with Sixteen Reader Group Identifiers bound to Single Access Credential

Table 7-15 NFC_UD_STANDARD_SIXTEEN_GROUPPIDENTIFIER_ONE_AC test identifiers

Parameter	Value
Test ID	NFC_UD_STANDARD_SIXTEEN_GROUPPIDENTIFIER_ONE_AC
PICS	Allow at least 16 reader_group_identifier per Access Credential
Applicability	M for User Device
Interface	NFC

Table 7-16 NFC_UD_STANDARD_SIXTEEN_GROUPPIDENTIFIER_ONE_AC test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK(i), reader_group_identifier(i), where i = 1, 2, 3, ...16.
TH (Reader)	Access Credential long term public key

Table 7-17 NFC_UD_STANDARD_SIXTEEN_GROUPPIDENTIFIER_ONE_AC test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then CONTINUE else FAIL.
5	Wait for at least 3 seconds.		
6	Repeat steps 1 through 5, for each of the sixteen reader_group_identifier.		Verify the following: 1. All iterations PASS 2. Same Access Credential long term public key is used in each iteration If all criteria above are met, then PASS else FAIL.

7.8 Expedited Standard Phase with Reader Certificate in LOAD_CERT with Chaining and incorrect Reader Cert signature

Table 7-18
NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_SIGNATURE test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_SIGNATURE
PICS	Verification of reader_Cert with the CA Public Key
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_SIGNATURE test pre-conditions are identical to NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 7-5.

Table 7-19
NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_SIGNATURE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Send LOAD_CERT command with fragmented reader_cert with chaining		
4		Send LOAD_CERT response	Verify the following: 1. SW = 9000h. 2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH1 with SW != 9000h routine (Table 5-4). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
6	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.9 Expedited Standard Phase with Reader Certificate in LOAD_CERT with Chaining and incorrect Reader Cert format

Table 7-20
NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_FORMAT test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_FORMAT
PICS	Verification of reader_Cert with the CA Public Key
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_FORMAT test pre-conditions are identical to NFC_UD_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 7-5.

Table 7-21
NFC_UD_NEG_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING_INCORRECT_FORMAT test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Send LOAD_CERT command with wrong value/length		
4		Send LOAD_CERT response	Verify the following: - SW != 9000h. - Response data field is empty. If all criteria are met, then CONTINUE else FAIL.
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-6).		If all criteria are met, then PASS else FAIL.

7.10 Step-Up Phase with Access Document

Table 7-22 NFC_UD_STEPUP_AD test identifiers

Parameter	Value
Test ID	NFC_UD_STEPUP_AD
PICS	Step-Up Phase AND Access Document storage and retrieval
Applicability	M for User Device
Interface	NFC

Table 7-23 NFC_UD_STEPUP_AD test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier, Access Document
TH (Reader)	Access Credential long term public key, IssuerKey_PubK

Table 7-24 NFC_UD_STEPUP_AD test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h. auth0_command_vendor_extension is present in AUTH0 command.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	If bit2 in signaling_bitmap in AUTH1 response is set to 1, then execute SELECT routine. Set AID = A000000909ACCE5502.		If all criteria are met, then CONTINUE else FAIL.
5	Request Access Document using DeviceRequest inside ENVELOPE command.		
6		Send Access Document in DeviceResponse inside ENVELOPE command response	Verify the following: Access Document is sent in ENVELOPE command response. If all criteria are met, then CONTINUE else FAIL.
7	one or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		
8	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.11 Step-Up Phase with Revocation Document

Table 7-25 NFC_UD_STEPUP_RD test identifiers

Parameter	Value
Test ID	NFC_UD_STEPUP_RD
PICS	Step-Up Phase AND Revocation Document storage and retrieval
Applicability	M for User Device
Interface	NFC

Table 7-26 NFC_UD_STEPUP_RD test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier, Revocation Document
TH (Reader)	Access Credential long term public key, IssuerKey_PubK

Table 7-27 NFC_UD_STEPUP_RD test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute test steps 1 through 4 in NFC_UD_STEPUP_AD (Table 7-24)		If all criteria are met, then CONTINUE else FAIL.
2	Request Revocation Document using DeviceRequest inside ENVELOPE command.		
3		Send Revocation Document in DeviceResponse inside ENVELOPE command response	Verify the following: Revocation Document is sent in ENVELOPE command response. If all criteria are met, then CONTINUE else FAIL.
7	one or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		
8	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.12 Step-Up Phase with Access Document and Revocation Document

Table 7-28 NFC_UD_STEPUP_AD_RD test identifiers

Parameter	Value
Test ID	NFC_UD_STEPUP_AD_RD
PICS	Step-Up Phase AND Revocation Document storage and retrieval AND Revocation Document storage and retrieval
Applicability	M for User Device
Interface	NFC

Table 7-29 NFC_UD_STEPUP_AD_RD test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier, Access Document, Revocation Document
TH (Reader)	Access Credential long term public key, IssuerKey_PubK

Table 7-30 NFC_UD_STEPUP_AD_RD test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute test steps 1 through 4 in NFC_UD_STEPUP_AD (Table 7-24)		If all criteria are met, then CONTINUE else FAIL.
2	Request Access Document and Revocation Document in a single DeviceRequest inside ENVELOPE command.		
3		Send Access Document and Revocation Document in DeviceResponse inside ENVELOPE command response	Verify the following: Access Document and Revocation Document is sent in ENVELOPE command response If all criteria are met, then CONTINUE else FAIL.
7	one or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		
8	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.13 SELECT Response with User Device Descriptor Tag (provisional)

Table 7-31 NFC_UD_SELECT_RESPONSE_UD_DESCRIPTOR_TAG test identifiers

Parameter	Value
Test ID	NFC_UD_SELECT_RESPONSE_UD_DESCRIPTOR_TAG
PICS	User Device Descriptor Tag
Applicability	M for User Device that supports sending User Device Descriptor Tag
Interface	NFC

NFC_UD_SELECT_RESPONSE_UD_DESCRIPTOR_TAG test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-32 NFC_UD_SELECT_RESPONSE_UD_DESCRIPTOR_TAG test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501		Verify the following in addition to all verification in SELECT routine: 1. User Device Descriptor TLV structure is present in SELECT response.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			2. User Device Descriptor TLV structure matches technical specification. If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.14 AUTH0 Response with Chaining

Table 7-33 NFC_UD_AUTH0_RESPONSE_CHAINING test identifiers

Parameter	Value
Test ID	NFC_UD_AUTH0_RESPONSE_CHAINING
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_AUTH0_RESPONSE_CHAINING test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-34 NFC_UD_AUTH0_RESPONSE_CHAINING test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h. AUTH0 command with single or multiple auth0_command_vendor_extension with chaining and le!=0		If all criteria are met, then CONTINUE else FAIL.
3	[Optional] Send one or more GET RESPONSE command.		
4		[Optional] Send one or more GET RESPONSE	Verify AUTH0 response is chained. If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
		command response.	
5	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.15 AUTH0 with Unknown Reader Identifier

Table 7-35 NFC_UD_NEG_AUTH0_UNKNOWN_READER_ID test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_UNKNOWN_READER_ID
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

Table 7-36 NFC_UD_NEG_AUTH0_UNKNOWN_READER_ID test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK
TH (Reader)	Access Credential long term public key

Table 7-37 NFC_UD_NEG_AUTH0_UNKNOWN_READER_ID test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h and reader_identifier to a random value		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW != 9000h routine (Table 5-4). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.16 AUTH0 with unsupported Protocol Version

Table 7-38 NFC_UD_NEG_AUTH0_UNSUPPORTED_PROTOCOL_VERSION test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_UNSUPPORTED_PROTOCOL_VERSION
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH0_UNSUPPORTED_PROTOCOL_VERSION test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-39 NFC_UD_NEG_AUTH0_UNSUPPORTED_PROTOCOL_VERSION test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
3	Send AUTH0 command with unsupported expedited_phase_protocol_version and command_parameters = 0h		
4		send AUTH0 response	Verify the following: 1. SW != 9000h. 2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.
5	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.17 AUTH0 with Extra Unknown TLV

Table 7-40 NFC_UD_NEG_AUTH0_EXTRA_TAG test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_EXTRA_TAG
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH0_EXTRA_TAG test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-41 NFC_UD_NEG_AUTH0_EXTRA_TAG test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h and add extra unknown tag in TLV. Extra tag can be randomly injected at any location in the command payload.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.18 AUTH0 with Wrong Value

Table 7-42 NFC_UD_NEG_AUTH0_WRONG_VALUE test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_WRONG_VALUE
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH0_WRONG_VALUE test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-43 NFC_UD_NEG_AUTH0_WRONG_VALUE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Send AUTH0 command with command_parameters = 0h and wrong value/length for tag.		
3		send AUTH0 response	Verify the following: 1. SW != 9000h. 2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.19 AUTH0 with Wrong P1 and P2

Table 7-44 NFC_UD_NEG_AUTH0_WRONG_P1P2 test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_WRONG_P1P2
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH0_WRONG_P1P2 test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-45 NFC_UD_NEG_AUTH0_WRONG_P1P2 test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Send AUTH0 command with command_parameters = 0h and wrong value of P1 and P2.		
3		send AUTH0 response	Verify the following: 1. SW != 9000h. 2. Response data field is empty. If all criteria are met, then CONTINUE else FAIL.
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.20 AUTH0 with Chaining Not Completed

Table 7-46 NFC_UD_NEG_AUTH0_CHAINING_NOT_COMPLETED test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_CHAINING_NOT_COMPLETED
PICS	Expedited-Standard Phase

Applicability	M for User Device
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NEG_AUTH0_CHAINING_NOT_COMPLETED test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-47 NFC_UD_NEG_AUTH0_CHAINING_NOT_COMPLETED test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Send AUTH0 command with command_parameters = 0h and with single auth0_command_vendor_extension with command chaining and skipping sending the last APDU in the chain.		
3		send AUTH0 response	
4	Execute AUTH1 with SW != 9000h routine (Table 5-4)		If all criteria are met, then CONTINUE else FAIL.
5	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL

7.21 AUTH0 with Different Cryptogram in Consecutive Expedited Fast Phase

Table 7-48 NFC_UD_NEG_AUTH0_DIFFERENT_CRYPTOGRAM_CONSECUTIVE_FAST test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH0_DIFFERENT_CRYPTOGRAM_CONSECUTIVE_FAST
PICS	Expedited-Fast Phase AND Cryptogram generation and validation
Applicability	M for User Device that support Expedited-Fast
Interface	NFC

User Device and Reader do not have any information about each other as a pre-condition to this test.

Table 7-49 NFC_UD_NEG_AUTH0_DIFFERENT_CRYPTOGRAM_CONSECUTIVE_FAST test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 1h.		If all criteria are met, then CONTINUE else FAIL.
3	Wait at least 3 seconds.		
4	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH0 routine (Table 5-2). Set command_parameters = 1h.		If all criteria are met, then CONTINUE else FAIL.
			Verify the following in AUTH0 command response: - Value of Tag 0x86h is different between step 2 and step 5 - Value of Tag 0x9Dh is different between step 2 and step 5 If all criteria are met, then PASS else FAIL.

7.22 AUTH1 with Wrong Reader Signature

Table 7-50 NFC_UD_NEG_AUTH1_WRONG_READER_SIGNATURE test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH1_WRONG_READER_SIGNATURE
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH1_WRONG_READER_SIGNATURE test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-51 NFC_UD_NEG_AUTH1_WRONG_READER_SIGNATURE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW != 9000h routine (Table 5-4) and send wrong reader signature in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.23 AUTH1 with Extra Tag

Table 7-52 NFC_UD_NEG_AUTH1_EXTRA_TAG test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH1_EXTRA_TAG
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH1_EXTRA_TAG test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-53 NFC_UD_NEG_AUTH1_EXTRA_TAG test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Add extra unknown tag in TLV. Extra tag can be randomly injected at any location in the command payload.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.24 AUTH1 with Wrong P1 and P2

Table 7-54 NFC_UD_NEG_AUTH1_WRONG_P1P2 test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH1_WRONG_P1P2
PICS	Expedited-Standard Phase
Applicability	M for User Device

Interface	NFC
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NFC_UD_NEG_AUTH1_WRONG_P1P2 test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-55 NFC_UD_NEG_AUTH1_WRONG_P1P2 test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW != 9000h routine (Table 5-4). Add wrong value of P1 and P2 in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.25 AUTH1 with Wrong Values

Table 7-56 NFC_UD_NEG_AUTH1_WRONG_VALUES test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH1_WRONG_VALUES
PICS	Expedited-Standard Phase
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH1_WRONG_VALUES test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-57 NFC_UD_NEG_AUTH1_WRONG_VALUES test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
3	Execute AUTH1 with SW != 9000h routine (Table 5-4) with wrong value/length for tags in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.26 AUTH1 with Incomplete Chaining

Table 7-58 NFC_UD_NEG_AUTH1_CHAINING_NOT_COMPLTED test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_AUTH1_CHAINING_NOT_COMPLTED
PICS	Command Chaining
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_AUTH1_CHAINING_NOT_COMPLTED test pre-conditions are identical to NFC_UD_STANDARD_NO_CERT test pre-conditions in Table 7-2.

Table 7-59 NFC_UD_NEG_AUTH1_CHAINING_NOT_COMPLTED test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Send AUTH1 command with chaining and skipping sending the last APDU in the chain.		
4		send AUTH1 response	Abort at this step, if AUTH1 response not sent. Otherwise, proceed to next step.
5	Send Select with AID = A000000909ACCE5502.		
6		Select Response	Verify the following: SW != 9000h. If all criteria are met, then CONTINUE else FAIL.
7	Execute CONTROL FLOW indicating transaction failure routine (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.27 EXCHANGE with Mailbox Read Request

Table 7-60 NFC_UD_EXCHANGE_READ_REQUEST test identifiers

Parameter	Value
Test ID	NFC_UD_EXCHANGE_READ_REQUEST
PICS	Mailbox - Read
Applicability	M for User Device, if bit 4 in signaling_bitmap in AUTH1 command response is set to 1
Interface	NFC

Table 7-61 NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier, non-zero mailbox populated with random data
TH (Reader)	Access Credential long term public key

Table 7-62 NFC_UD_EXCHANGE_READ_REQUEST test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command multiple times with multiple READ requests from mailbox		
5		Send Exchange command response	Verify the following: Read requests return random data in mailbox. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.28 EXCHANGE with Mailbox Write Request

Table 7-63 NFC_UD_EXCHANGE_WRITE_REQUEST test identifiers

Parameter	Value
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Test ID	NFC_UD_EXCHANGE_WRITE_REQUEST
PICS	Mailbox - Write
Applicability	M for User Device, if bit 5 in signaling_bitmap in AUTH1 command response is set to 1
Interface	NFC

NFC_UD_EXCHANGE_WRITE_REQUEST test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-64 NFC_UD_EXCHANGE_WRITE_REQUEST test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command multiple times with multiple WRITE requests to mailbox (atomic session = TRUE).		
5		Send Exchange command response	Verify the following: Write requests do not fail. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
6	Send EXCHANGE command with atomic session = FALSE and random requests		
7		Send EXCHANGE command response	Verify the following: If Read Request is present, non-updated data is returned. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
8	Send EXCHANGE command to read data written to the mailbox		
9		Send EXCHANGE command response	Verify the following: Read request should return final data in the mailbox after closing of atomic session B1 and B2 are both 00h.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			If all criteria are met, then CONTINUE else FAIL.
10	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.29 EXCHANGE with Set Request

Table 7-65 NFC_UD_EXCHANGE_SET_REQUEST test identifiers

Parameter	Value
Test ID	NFC_UD_EXCHANGE_SET_REQUEST
PICS	Mailbox
Applicability	M for User Device, if bit 5 in signaling_bitmap in AUTH1command response is set to 1
Interface	NFC

NFC_UD_EXCHANGE_SET_REQUEST test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-66 NFC_UD_EXCHANGE_SET_REQUEST test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command multiple times with multiple SET requests to mailbox (atomic session = TRUE).		
5		Send Exchange command response	Verify the following: Write requests do not fail. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
6	Send EXCHANGE command with atomic session = FALSE and random requests		
7		Send EXCHANGE command response	Verify the following: If Read Request is present, non-updated data is returned.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
8	Send EXCHANGE command to read data written to the mailbox		
9		Send EXCHANGE command response	Verify the following: Read request should return final data in the mailbox after closing of atomic session. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
10	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.30 EXCHANGE with Chaining

Table 7-67 NFC_UD_EXCHANGE_WITH_CHAINING test identifiers

Parameter	Value
Test ID	NFC_UD_EXCHANGE_WITH_CHAINING
PICS	EXCHANGE command
Applicability	M for User Device
Interface	NFC

NFC_UD_EXCHANGE_WITH_CHAINING test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-68 NFC_UD_EXCHANGE_WITH_CHAINING test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
7	Send EXCHANGE command multiple times with chaining with Read/Write		

Steps	TH (Reader)	DUT (User Device)	Verification at TH
	requests and le!=0 (atomic session = TRUE)		
8		[Optional] Send Exchange command response	Send GET RESPONSE multiple times to retrieve complete response
4	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.31 EXCHANGE with Extended Length

Table 7-69 NFC_UD_EXCHANGE_WITH_EXTENDED_LENGTH test identifiers

Parameter	Value
Test ID	NFC_UD_EXCHANGE_WITH_EXTENDED_LENGTH
PICS	EXCHANGE command AND Extended Length
Applicability	M for User Device that supports Extended Length
Interface	NFC

NFC_UD_EXCHANGE_WITH_EXTENDED_LENGTH test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-70 NFC_UD_EXCHANGE_WITH_EXTENDED_LENGTH test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command with extended length APDU and large Read request		
5		Send Exchange command response	Verify the following: Reading Mailbox should not fail. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.32 EXCHANGE with Extra Tag

Table 7-71 NFC_UD_NEG_EXCHANGE_WITH_EXTRA_TAG test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_EXCHANGE_WITH_EXTRA_TAG
PICS	EXCHANGE command
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_EXCHANGE_WITH_EXTRA_TAG test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-72 NFC_UD_NEG_EXCHANGE_WITH_EXTRA_TAG test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command with extra tag in encrypted payload		
5		Send Exchange command response	Verify the following: All requests should pass. B1 and B2 are both 00h. If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 5-5).		If all criteria are met, then PASS else FAIL.

7.33 EXCHANGE with Mailbox Out of Bounds

Table 7-73 NFC_UD_NEG_EXCHANGE_MAILBOX_OUT_OF_BOUNDS test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_EXCHANGE_MAILBOX_OUT_OF_BOUNDS
PICS	Mailbox
Applicability	M for User Device

Interface	NFC
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NFC_UD_NEG_EXCHANGE_MAILBOX_OUT_OF_BOUNDS test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-74 NFC_UD_NEG_EXCHANGE_MAILBOX_OUT_OF_BOUNDS test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1).. Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command with mailbox request with offset + length > mailbox size		
5		Send Exchange command response	Verify the following: Exchange command response SW = 9000h and response payload has 0x0002 B1 B2, where B1 and B2 are implementation specific. If all criteria are met, then CONTINUE else FAIL.
6	Execute CONTROL FLOW indicating transaction failure (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.34 EXCHANGE with Wrong Length

Table 7-75 NFC_UD_NEG_EXCHANGE_WITH_WRONG_LENGTH test identifiers

Parameter	Value
Test ID	NFC_UD_NEG_EXCHANGE_WITH_WRONG_LENGTH
PICS	EXCHANGE command
Applicability	M for User Device
Interface	NFC

NFC_UD_NEG_EXCHANGE_WITH_WRONG_LENGTH test pre-conditions are identical to NFC_UD_EXCHANGE_READ_REQUEST test pre-conditions in Table 7-61.

Table 7-76 NFC_UD_NEG_EXCHANGE_WITH_WRONG_LENGTH test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute SELECT routine (Table 5-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 5-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 with SW = 9000h routine (Table 5-3). Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4	Send EXCHANGE command with wrong length/value for tag		
5		Send Exchange command response	Verify the following: Exchange command response SW = 9000h and response payload has 0x0002 B1 B2, where B1 and B2 are implementation specific. If all criteria are met, then CONTINUE else FAIL.
6	Execute CONTROL FLOW indicating transaction failure (Table 5-7).		If all criteria are met, then PASS else FAIL.

7.35 BLE+UWB Flow with Expedited Standard Phase

Table 7-77 BLEUWB_UD_EXPEDITED_STANDARD_PHASE test identifiers

Parameter	Value
Test ID	BLEUWB_UD_EXPEDITED_STANDARD_PHASE
PICS	BLE + UWB Flow AND UWB ranging AND UWB Time Synchronization AND Expedited-Standard Phase
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

Table 7-78 BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier, GRK
TH (Reader)	Access Credential long term public key, GRK
NOTE 1: The TH (Reader) and the DUT (User Device) are in very close proximity (e.g., 1 m and line-of-sight). NOTE 2: TH is in secured state as a pre-condition.	

Table 7-79 BLEUWB_UD_EXPEDITED_STANDARD_PHASE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		If all criteria are met, then CONTINUE else FAIL.
2		Send Time Sync Message ID.	Verify the following: 1. Confirm Time Sync Message ID is received by TH 2. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
3		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID.	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
4	Execute BLE+UWB ranging session setup routine (Table 5-9).		If all criteria are met, then CONTINUE else FAIL.
5	Allow N (e.g., 3 seconds) for UWB ranging to occur.		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
6	Reader Status Changed Message ID carrying State Attribute ID is sent.		Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

7.36 BLE+UWB Flow with Expedited Fast Phase

Table 7-80 BLEUWB_UD_EXPEDITED_FAST_PHASE test identifiers

Parameter	Value
Test ID	BLEUWB_UD_EXPEDITED_FAST_PHASE
PICS	BLE + UWB Flow AND UWB ranging AND UWB Time Synchronization AND Expedited-Fast Phase
Applicability	M for User Device that supports BLE + UWB Flow and Expedited-Fast phase
Interface	BLE

BLEUWB_UD_EXPEDITED_FAST_PHASE test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-81 BLEUWB_UD_EXPEDITED_FAST_PHASE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		If all criteria are met, then CONTINUE else FAIL.
2	BLE teardown		
3	Send Bluetooth LE advertisement		
4		Establish L2CAP connection	
5		Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501	Verify the following: Format of Initiate Access Protocol Message ID matches specification. If all criteria are met, then CONTINUE else FAIL.
6	Send AUTH0 command command_parameters = 1h authentication_policy = 01h (User Device)		
7		send AUTH0 response	Verify the following: Format of AUTH0 response matches specification. If all criteria are met, then CONTINUE else FAIL.
8	Send EXCHANGE command		Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.
9		Send EXCHANGE response	
10	Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID		Verify the following: Ensure reader status is secured. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
11		Send Time Sync Message ID	Verify the following: 1. Confirm Time Sync Message ID is received by TH 2. Format of message matches technical specification.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			If all criteria are met, then CONTINUE else FAIL.
12		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
13	Execute BLE+UWB ranging session setup routine (Table 5-9).		If all criteria are met, then CONTINUE else FAIL.
14	Allow N (e.g., 3 seconds) for UWB ranging to occur.		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
15	Reader Status Changed Message ID carrying State Attribute ID is sent.		Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

7.37 BLE+UWB Flow with Step-Up Phase

Table 7-82 BLEUWB_UD_STEPUP_PHASE test steps

Parameter	Value
Test ID	BLEUWB_UD_STEPUP_PHASE
PICS	BLE + UWB Flow AND UWB ranging AND UWB Time Synchronization AND Step-Up Phase
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

Table 7-83 BLEUWB_UD_STEPUP_PHASE test pre-conditions

Provision onto	Remarks
DUT (User Device)	reader_PubK, reader_group_identifier, GRK, Access Document
TH (Reader)	Access Credential long term public key, GRK, IssuerKey_PubK
NOTE 1: The TH (Reader) and the DUT (User Device) are in very close proximity (e.g., 1 m and line-of-sight).	
NOTE 2: TH is in secured state as a pre-condition.	

Table 7-84 BLEUWB_UD_STEPUP_PHASE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement		
2		Establish L2CAP connection	
3		Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501	Verify the following: Format of Initiate Access Protocol Message ID matches specification. If all criteria are met, then CONTINUE else FAIL.
4	Send AUTH0 command command_parameters = 0h authentication_policy = 01h (User Device)		
5		send AUTH0 response	Verify the following: Format of AUTH0 response matches specification. If all criteria are met, then CONTINUE else FAIL.
6	Send AUTH1 command and reader_cert is absent		
7		send AUTH1 response	Verify the following: Format of AUTH1 response matches specification. If all criteria are met, then CONTINUE else FAIL.
8	Send EXCHANGE command		Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.
9		Send EXCHANGE response	
10	Request Access Document using DeviceRequest inside ENVELOPE command		
11		Send Access Document in DeviceResponse inside ENVELOPE command response	Verify the following: Access Document is sent in ENVELOPE command response. If all criteria are met, then CONTINUE else FAIL

Steps	TH (Reader)	DUT (User Device)	Verification at TH
12	One or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		
13	Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID		Verify the following: Ensure reader status is secured. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
14		Send Time Sync Message ID	Verify the following: 1. Confirm Time Sync Message ID is received by TH 2. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
15		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
16	Execute BLE+UWB ranging session setup routine (Table 5-9).		
17	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
18	Reader Status Changed Message ID carrying State Attribute ID is sent		Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

7.38 BLE+UWB Flow with UWB Ranging Suspend

Table 7-85 BLEUWB_UD_RANGING_SUSPEND test identifiers

Parameter	Value
Test ID	BLEUWB_UD_RANGING_SUSPEND
PICS	BLE + UWB Flow AND UWB ranging suspend
Applicability	M for User Device that supports BLE + UWB Flow

Interface	BLE
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BLEUWB_UD_RANGING_SUSPEND test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78 with the exception: The TH (Reader) and the DUT (User Device) are in close proximity (e.g., 5 m and line-of-sight).

Table 7-86 BLEUWB_UD_RANGING_SUSPEND test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		If all criteria are met, then CONTINUE else FAIL.
2		Send Time Sync Message ID	Verify the following: 1. Confirm Time Sync Message ID is received by TH 2. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
3		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
4	Execute BLE+UWB ranging session setup routine (Table 5-9).		If all criteria are met, then CONTINUE else FAIL.
5	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
6	Send Ranging Session Suspend Request with correct UWB Session Identifier		
7		Send Ranging Session Suspend Response	Verify the following 1. this message is sent. 2. format of Ranging Session Suspend Response matches technical specification. 3. No UWB packets are received over UWB transport a short time (e.g., up to 3 seconds) after receiving Ranging Session Suspend Response. If all criteria are met, then PASS else FAIL. The status can either value 0 or 1.

7.39 BLE+UWB Flow with UWB Ranging Resume

Table 7-87 BLEUWB_UD_RANGING_RESUME test identifiers

Parameter	Value
Test ID	BLEUWB_UD_RANGING_RESUME
PICS	BLE + UWB Flow AND UWB ranging resume
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_RANGING_RESUME test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78 with the exception: The TH (Reader) and the DUT (User Device) are in close proximity (e.g., 5 m and line-of-sight)..

Table 7-88 BLEUWB_UD_RANGING_RESUME test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		If all criteria are met, then CONTINUE else FAIL.
2		Send Time Sync Message ID	Verify the following: - Confirm Time Sync Message ID is received by TH - Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
3		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
4	Execute BLE+UWB ranging session setup routine (Table 5-9).		If all criteria are met, then CONTINUE else FAIL.
5	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
6	Send Ranging Message ID carrying Ranging Session Suspended Attribute ID		Verify the following: No UWB packets are received over UWB transport a short time (e.g., up to 3 seconds) after sending Ranging Message ID carrying Ranging Session Suspended Attribute ID.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
7	1 second after previous step, send Ranging Session Resume Request Message		
8		Send Ranging Session Resume Response or Ranging Message ID carrying Initiate Ranging Session Resume Later Attribute ID	Verify the following: 1. Ranging Session Resume Response or Ranging Message ID carrying Initiate Ranging Session Resume Later Attribute ID is sent. 2. Format of the message matches technical specification. If all criteria are met, then PASS else FAIL.

7.40 BLE+UWB Flow with User Device Descriptor Tag (provisional)

Parameter	Value
Test ID	BLEUWB_UD_UD_DESCRIPTOR_TAG
PICS	BLE + UWB Flow AND User Device Descriptor Tag
Applicability	M for User Device that supports BLE + UWB Flow and that supports sending User Device Descriptor Tag
Interface	BLE

BLEUWB_UD_UD_DESCRIPTOR_TAG test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-89 BLEUWB_UD_UD_DESCRIPTOR_TAG test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		Verify the following in Initiate Access Protocol Message ID in addition to other verifications in BLE+UWB Aliro Access Protocol routine. 1. Proprietary Information Attribute ID is present. 2. The format of Proprietary Information ID matches the technical specification. 3. User Device Descriptor TLV structure is present in Proprietary Information Attribute ID. If all criteria are met, then PASS else FAIL.

7.41 BLE+UWB Flow with wrong advertisement format

Table 7-90 BLEUWB_UD_NEG_WRONG_ADV test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_WRONG_ADV
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_WRONG_ADV test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-91 BLEUWB_UD_NEG_WRONG_ADV test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement with bits 6 and 7 in byte 7, each set to 0 for 30 seconds.		
2		No BLE connection initiated by the User Device.	Verify the following: DUT (User Device) does not establish BLE connection with the TH (Reader). If all criteria are met, then PASS else FAIL.

7.42 BLE+UWB Flow with Failed L2CAP

Table 7-92 BLEUWB_UD_NEG_FAILED_L2CAP test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_FAILED_L2CAP
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_FAILED_L2CAP test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-93 BLEUWB_UD_NEG_FAILED_L2CAP test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement.		
2	Reader sends wrong Supported Aliro Ble UWB Protocol Version.	Establish L2CAP connection fails.	Verify the following:

Steps	TH (Reader)	DUT (User Device)	Verification at TH
			L2CAP establishment fails. If all criteria are met, then PASS else FAIL.

7.43 BLE+UWB Flow with timeout before AUTH0

Table 7-94 BLEUWB_UD_NEG_TIMEOUT_BEFORE_AUTH0 test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_TIMEOUT_BEFORE_AUTH0
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_TIMEOUT_BEFORE_AUTH0 test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-95 BLEUWB_UD_NEG_TIMEOUT_BEFORE_AUTH0 test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement.		
2		Establish L2CAP connection.	
3		Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501.	Verify the following: Format of Initiate Access Protocol Message ID matches specification. If all criteria are met, then CONTINUE else FAIL.
4	Do not send AUTH0 command.		
5		BLE teardown.	Verify the following: BLE teardown is initiated by the DUT (User Device). If all criteria are met, then PASS else FAIL.

7.44 BLE+UWB Flow with Timeout Extension

Table 7-96 BLEUWB_UD_TIMEOUT_EXTENSION test identifiers

Parameter	Value
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Test ID	BLEUWB_UD_TIMEOUT_EXTENSION
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_TIMEOUT_EXTENSION test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-97 BLEUWB_UD_TIMEOUT_EXTENSION test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Send Bluetooth LE advertisement.		
2		Establish L2CAP connection.	
3		Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501.	Verify the following: Format of Initiate Access Protocol Message ID matches specification. If all criteria are met, then CONTINUE else FAIL.
4	Send Event Message ID carrying Busy Attribute ID at 1 second after receiving Initiate Access Protocol Message ID.		
5	1 second of sending Event Busy Attribute ID, execute AUTH0 routine. Set command_parameters = 0h and authentication_policy = 01h (User Device).		If all criteria are met, then CONTINUE else FAIL.
6	Execute AUTH1 routine with SW = 9000h. Reader_cert is not present in AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
7	Send EXCHANGE command		Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.
8		Send EXCHANGE response	
9	Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID		Verify the following: Ensure reader status is secured or unsecured. Format of message matches technical specification. If all criteria are met, then PASS else FAIL.

7.45 BLE+UWB Flow with URSK Not Found

Table 7-98 BLEUWB_UD_NEG_URSK_NOT_FOUND test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_URSK_NOT_FOUND
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_URSK_NOT_FOUND test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-99 BLEUWB_UD_NEG_URSK_NOT_FOUND test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8). When sending EXCHANGE command do not include Tag 0x98 in it.		If all criteria are met, then CONTINUE else FAIL.
2		Send Event Message ID carrying General Error Attribute ID with URSK unavailable	Verify the following: Format of this message matches the specification. If all criteria are met, then PASS else FAIL.
3	BLE teardown		

7.46 BLE+UWB Flow with M1 Message Mismatch Parameter

Table 7-100 BLEUWB_UD_NEG_M1_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_M1_MISMATCH_PARAMETER
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_M1_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-101 BLEUWB_UD_NEG_M1_MISMATCH_PARAMETER test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (Reader)	DUT (User Device)	Verification at TH
2		Send Time Sync Message ID	Verify the following: 1. Confirm Time Sync Message ID is received by TH 2. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
3		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
5	Send Ranging Session Setup M1 Message ID without UWB Config ID		
6		Send Event Message ID carrying General Error Attribute ID indicating Wrong Parameters	Verify the following: Format of Event Message ID. General Error Attribute ID indicates Wrong Parameters. If all criteria are met, then PASS else FAIL.
7	BLE teardown		

7.47 BLE+UWB Flow with M3 Message Mismatch Parameter

Table 7-102 BLEUWB_UD_NEG_M3_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_M3_MISMATCH_PARAMETER
PICS	BLE + UWB Flow
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_M3_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-103 BLEUWB_UD_NEG_M3_MISMATCH_PARAMETER test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 5-8).		If all criteria are met, then CONTINUE else FAIL.
2		Send Time Sync Message ID	Verify the following: 1. Confirm Time Sync Message ID is received by TH 2. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
3		Send Ranging Message ID carrying Initiate Ranging Session Attribute ID	Verify the following: Format of this message matches the specification. If all criteria are met, then CONTINUE else FAIL.
4	Send Ranging Session Setup M1 Message ID		
5		Send Ranging Session Setup M2 Message ID	
6	Send Ranging Session Setup M3 Message ID without RAN Multiplier		
7		Send Event Message ID carrying General Error Attribute ID indicating Wrong Parameters	Verify the following: Format of Event Message ID. General Error Attribute ID indicates Wrong Parameters. If all criteria are met, then PASS else FAIL.
8	BLE teardown		

7.48 BLE+UWB Flow with Suspend Request Mismatch Parameter

Table 7-104 BLEUWB_UD_NEG_SUSPEND_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_SUSPEND_MISMATCH_PARAMETER
PICS	BLE + UWB Flow AND UWB Ranging Suspend
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_SUSPEND_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-105 BLEUWB_UD_NEG_SUSPEND_MISMATCH_PARAMETER test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLEUWB_UD_RANGING_SUSPEND test steps (Table 7-86). Send Ranging Session Suspend Request with an incorrect UWB Session Identifier	Send Event Message ID carrying General Error Attribute ID indicating Wrong Parameters	Verify the following: Format of Event Message ID. General Error Attribute ID indicates Wrong Parameters. If all criteria are met, then PASS else FAIL.
2	BLE teardown		

7.49 BLE+UWB Flow with Resume Request Mismatch Parameter

Table 7-106 BLEUWB_UD_NEG_RESUME_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_UD_NEG_RESUME_MISMATCH_PARAMETER
PICS	BLE + UWB Flow AND UWB Ranging Resume
Applicability	M for User Device that supports BLE + UWB Flow
Interface	BLE

BLEUWB_UD_NEG_RESUME_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-107 BLEUWB_UD_NEG_RESUME_MISMATCH_PARAMETER test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLEUWB_UD_RANGING_RESUME test steps (Table 7-88). Send Ranging Session Resume Request with an incorrect UWB Session Identifier.	Send Event Message ID carrying General Error Attribute ID indicating Wrong Parameters.	Verify the following: Format of Event Message ID. General Error Attribute ID indicates Wrong Parameters. If all criteria are met, then PASS else FAIL.
2	BLE teardown		

7.50 BLE-Only Flow with Expedited Standard Phase (provisional)

Table 7-108 BLERKE_UD_EXPEDITED_STANDARD_PHASE test identifiers

Parameter	Value
Test ID	BLERKE_UD_EXPEDITED_STANDARD_PHASE
PICS	BLE-Only Flow AND Explicit Reader Selection
Applicability	M for User Device that supports BLE-Only Flow
Interface	BLE

BLERKE_UD_EXPEDITED_STANDARD_PHASE test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-109 BLERKE_UD_EXPEDITED_STANDARD_PHASE test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE-Only Aliro Access Protocol routine (Table 5-10).		If all criteria are met, then CONTINUE else FAIL.
2		Send RKE Request Message ID carrying Action Attribute ID	Verify the following: Format of Message ID matches the technical specification. If all criteria are met, then CONTINUE else FAIL.
3	Send Reader Status Changed Message ID carrying State Attribute ID.		Verify the following: Format of Reader Status Changed Message ID carrying State Attribute ID matches technical specification. If all criteria are met, then PASS else FAIL.

7.51 BLE-Only Flow with User Device Descriptor Tag (provisional)

Table 7-110 BLERKE_UD_UD_DESCRIPTOR_TAG test identifiers

Parameter	Value
Test ID	BLERKE_UD_UD_DESCRIPTOR_TAG
PICS	BLE-Only Flow
Applicability	M for User Device that supports BLE-Only Flow and that supports sending User Device Descriptor Tag
Interface	BLE

BLERKE_UD_UD_DESCRIPTOR_TAG test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

Table 7-111 BLERKE_UD_UD_DESCRIPTOR_TAG test steps

Steps	TH (Reader)	DUT (User Device)	Verification at TH
1	Execute BLE-Only Aliro Access Protocol routine (Table 5-10).		Verify the following in Initiate Access Protocol RKE Message ID, in addition to verifying BLE-Only Aliro Access Protocol routine: 1. Proprietary Information Attribute ID is present 2. The format of Proprietary Information ID is matching technical specification 3. User Device Descriptor TLV structure is present in Proprietary Information Attribute ID If all criteria are met, then PASS else FAIL.

7.52 BLE-Only Flow with Failed L2CAP (provisional)

Table 7-112 BLERKE_UD_NEG_FAILED_L2CAP test identifiers

Parameter	Value
Test ID	BLERKE_UD_NEG_FAILED_L2CAP
PICS	BLE-Only Flow
Applicability	M for User Device that supports BLE-Only Flow
Interface	BLE

BLERKE_UD_NEG_FAILED_L2CAP test pre-conditions are identical to BLEUWB_UD_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 7-78.

BLERKE_UD_NEG_FAILED_L2CAP test steps are identical to BLEUWB_UD_NEG_FAILED_L2CAP test steps in Table 7-93.

8 Reader Under Test Conformance Tests

8.1 Expedited Standard Phase without Reader Certificate

Table 8-1 NFC_RDR_STANDARD_NO_CERT test identifiers

Parameter	Value
Test ID	NFC_RDR_STANDARD_NO_CERT
PICS	Expedited-Standard Phase AND Reader signature generation and validation using reader_PubK AND Device signature generation and validation AND AUTH1 command parameter
Applicability	M for Reader
Interface	NFC

Table 8-2 NFC_RDR_STANDARD_NO_CERT test pre-conditions

Provision onto	Remarks
DUT (Reader)	Access Credential long term public key
TH (User Device)	reader_PubK, reader_group_identifier

Table 8-3 NFC_RDR_STANDARD_NO_CERT test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.2 Expedited Standard Phase with Reader Certificate in LOAD_CERT with APDU Chaining

Table 8-4 NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test identifiers

Parameter	Value
Test ID	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING
PICS	Expedited-Standard Phase AND

	Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Device signature generation and validation AND Presentation and validation of the reader_Cert in LOAD_CERT command AND Command chaining AND AUTH1 command parameter
Applicability	M for Reader
Interface	NFC

Table 8-5 NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions

Provision onto	Remarks
DUT (Reader)	Access Credential long term public key
TH (User Device)	Reader System Issuer CA public key, reader_group_identifier

Table 8-6 NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3		Send LOAD_CERT command with fragmented reader_cert with chaining	Verify the following: reader_cert with chaining sent. If all criteria are met, then CONTINUE else FAIL.
4	Send LOAD_CERT Response		
5	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.3 Expedited Standard Phase with Reader Cert in LOAD_CERT without APDU Chaining

Table 8-7 NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING test identifiers

Parameter	Value
Test ID	NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING
PICS	Expedited-Standard Phase AND

	Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Device signature generation and validation AND Presentation and validation of the reader_Cert in LOAD_CERT command AND Extended length AND AUTH1 command parameter
Applicability	M for Reader, if it supports Extended length APDUs
Interface	NFC

NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING test pre-conditions are identical to NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 8-5.

Table 8-8 NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_NO_CHAINING test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3		Send LOAD_CERT command with reader_cert and no chaining	Verify the following: Reader_cert sent without chaining. If all criteria are met, then CONTINUE else FAIL.
4	Send LOAD_CERT Response		
5	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
6	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.4 Expedited Standard Phase with Reader Cert in AUTH1 with APDU Chaining

Table 8-9 NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING test identifiers

Parameter	Value
Test ID	NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING
PICS	Expedited-Standard Phase AND Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Device signature generation and validation AND Presentation and validation of the reader_Cert in AUTH1 command AND Command chaining AND

	AUTH1 command parameter
Applicability	M for Reader
Interface	NFC

NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING test pre-conditions are identical to NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 8-5.

Table 8-10 NFC_RDR_STANDARD_CERT_IN_AUTH1_WITH_CHAINING test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3). fragmented reader_cert with chaining over multiple APDUs.		Verify AUTH1 routine criteria and the following: Reader_cert chaining. If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.5 Expedited Standard Phase with Reader Cert in AUTH1 without Chaining

Table 8-11 NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING test steps

Parameter	Value
Test ID	NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING
PICS	Expedited-Standard Phase AND Reader signature generation and validation using intermediate_reader_PubK (from reader_Cert) AND Device signature generation and validation AND Presentation and validation of the reader_Cert in AUTH1 command AND Extended length AND AUTH1 command parameter
Applicability	M for Reader, if it supports Extended length APDU and supports sending reader_Cert in AUTH1 command
Interface	NFC

NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING test pre-conditions are identical to NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 8-5.

Table 8-12 NFC_RDR_STANDARD_CERT_IN_AUTH1_NO_CHAINING test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3). reader_cert with no chaining over multiple APDUs.		Verify AUTH1 routine criteria and the following: Reader_cert is not chained. If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.6 Expedited Fast Phase

Table 8-13 NFC_RDR_FAST test identifiers

Parameter	Value
Test ID	NFC_RDR_FAST
PICS	Expedited-fast AND Cryptogram generation and validation
Applicability	M for Reader that supports Expedited-Fast Phase
Interface	NFC

NFC_RDR_FAST test pre-conditions are identical to
NFC_RDR_STANDARD_CERT_IN_LOAD_CERT_WITH_CHAINING test pre-conditions in Table 8-5.

Table 8-14 NFC_RDR_FAST test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute NFC_RDR_STANDARD_NO_CERT test steps (Table 8-3)		If all criteria are met, then CONTINUE else FAIL.
2	Wait for at least 3 seconds.		
3	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
4	Execute AUTH0 routine (Table 6-2). Use reader_identifier value from step 1. command_parameters = 1h		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
5	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.7 Step-Up Phase with Minimal Access Document with Key Identifier

Table 8-15 NFC_RDR_STEPUP_AD_KEY_ID test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_KEY_ID
PICS	Step-Up Phase AND Step-Up AID Select AND Access document storage and retrieval
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

Table 8-16 NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions

Provision onto	Remarks
DUT (Reader)	IssuerKey_PubK
TH (User Device)	reader_PubK, reader_group_identifier, Access Document

Table 8-17 NFC_RDR_STEPUP_AD_KEY_ID test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
4		Send SELECT command AID = A000000909ACCE5502, if Step-Up AID is required is indicated in signaling_bitmap in AUTH1 response	If all criteria are met, then CONTINUE else FAIL.
5	Send SELECT response		
6		Request Access Document using	Verify the following:

Steps	TH (User Device)	DUT (Reader)	Verification at TH
		DeviceRequest inside ENVELOPE command.	Access Document is requested in ENVELOPE command response. If all criteria are met, then CONTINUE else FAIL.
7	Send appropriate Access Document in DeviceResponse inside ENVELOPE command response		
8	One or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		
9	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.8 Step-Up Phase with Minimal Access Document with Issuer Certificate

Table 8-18 NFC_RDR_STEPUP_AD_ISSUER_CERT test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_ISSUER_CERT
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

Table 8-19 NFC_RDR_STEPUP_AD_ISSUER_CERT test pre-conditions

Provision onto	Remarks
DUT (Reader)	Credential Issuer CA Certificate or Credential Issuer CA public key
TH (User Device)	reader_PubK, reader_group_identifier, Access Document

Repeat Table 8-17 with Access Document with Issuer Certificate.

8.9 Step-Up Phase with Minimal Access Document with both Issuer Certificate and Key ID

Table 8-20 NFC_RDR_STEPUP_AD_ISSUER_CERT_KEY_ID test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_ISSUER_CERT_KEY_ID
PICS	Step-Up Phase AND Access Document processing

Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

Table 8-21 NFC_RDR_STEPUP_AD_ISSUER_CERT_KEY_ID test pre-conditions

Provision onto	Remarks
DUT (Reader)	IssuerKey_PubK, Credential Issuer CA Certificate or Credential Issuer CA public key
TH (User Device)	reader_PubK, reader_group_identifier, Access Document

Repeat Table 8-17 with Access Document with Issuer Certificate and Key Identifier.

8.10 Step-Up Phase with Access Document with AccessRule

Table 8-22 NFC_RDR_STEPUP_AD_ACCESS_RULE test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_ACCESS_RULE
PICS	Step-Up Phase AND Access Data element verification – Access Rules
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_STEPUP_AD_ACCESS_RULE test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with AccessRule.

8.11 Step-Up Phase with Access Document with AccessRule using Schedules

Table 8-23 NFC_RDR_STEPUP_AD_ACCESS_RULE_SCHEDULES test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_ACCESS_RULE_SCHEDULES
PICS	Step-Up Phase AND Access Data element verification – Access Rules AND Schedules
Applicability	M for Reader that supports Step-Up Phase and that supports schedules
Interface	NFC

NFC_RDR_STEPUP_AD_ACCESS_RULE_SCHEDULES test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with AccessRule with schedules.

8.12 Step-Up Phase with Access Document with Unknown NonAccessExtension

Table 8-24 NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

Table 8-25 NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION test pre-conditions

Provision onto	Remarks
DUT (Reader)	IssuerKey_PubK, does not parse Extensions from Vendor_RegisteredID 000001
TH (User Device)	reader_PubK, reader_group_identifier, Access Document

Repeat Table 8-17 with Access Document with unknown nonAccessExtension.

8.13 Step-Up Phase with Access Document with Unknown Non-Critical AccessExtension

Table 8-26 NFC_RDR_STEPUP_AD_UNKNOWN_NON_CRITICAL_ACCESS_EXTENSION test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_AD_UNKNOWN_NON_CRITICAL_ACCESS_EXTENSION
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_STEPUP_AD_UNKNOWN_NON_CRITICAL_ACCESS_EXTENSION test pre-conditions are identical to NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION test pre-conditions in Table 8-25.

Repeat Table 8-17 with Access Document with unknown non-critical AccessExtension.

8.14 Step-Up Phase with Access Document with No Issuer Certificate or Key ID

Table 8-27 NFC_RDR_NEG_STEPUP_AD_NO_ISSUER_CERT_NO_KEY_ID test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_NO_ISSUER_CERT_NO_KEY_ID
PICS	Step-Up Phase AND

	Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_NO_ISSUER_CERT_NO_KEY_ID test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with no issuer certificate and no key identifier. EXCHANGE command indicates transaction failure for test to pass.

8.15 Step-Up Phase with Access Document with Issuer Certificate with Invalid Signature

Table 8-28 NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_INVALID_SIGNATURE test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_INVALID_SIGNATURE
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_INVALID_SIGNATURE test pre-conditions are identical to NFC_RDR_STEPUP_AD_ISSUER_CERT test pre-conditions in Table 8-19.

Repeat Table 8-17 with Access Document with issuer invalid signature. EXCHANGE command indicates transaction failure for test to pass.

8.16 Step-Up Phase with Access Document with Expired Issuer Certificate

Table 8-29 NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_EXPIRED test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_EXPIRED
PICS	Step-Up Phase AND Access Document verification – Validate time-based elements
Applicability	M for Reader that supports time concept and Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_ISSUER_CERT_EXPIRED test pre-conditions are identical to NFC_RDR_STEPUP_AD_ISSUER_CERT test pre-conditions in Table 8-19.

Repeat Table 8-17 with Access Document with expired issuer certificate. EXCHANGE command indicates transaction failure for test to pass.

8.17 Step-Up Phase with Access Document with Invalid Signature in IssuerAuth

Table 8-30 NFC_RDR_NEG_STEPUP_AD_INVALID_SIGNATURE_ISSUER_AUTH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_INVALID_SIGNATURE_ISSUER_AUTH
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_INVALID_SIGNATURE_ISSUER_AUTH test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with invalid signature in IssuerAuth. EXCHANGE command indicates transaction failure for test to pass.

8.18 Step-Up Phase with Access Document with Invalid Hash in IssuerAuth

Table 8-31 NFC_RDR_NEG_STEPUP_AD_INVALID_HASH_ISSUER_AUTH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_INVALID_HASH_ISSUER_AUTH
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_INVALID_HASH_ISSUER_AUTH test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with invalid hash in IssuerAuth. EXCHANGE command indicates transaction failure for test to pass.

8.19 Step-Up Phase with Access Document with Expired IssuerAuth

Table 8-32 NFC_RDR_NEG_STEPUP_AD_EXPIRED_ISSUER_AUTH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_EXPIRED_ISSUER_AUTH
PICS	Step-Up Phase AND Access Document verification – Validate time-based elements
Applicability	M for Reader that supports time concept and Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_EXPIRED_ISSUER_AUTH test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with expired IssuerAuth. EXCHANGE command indicates transaction failure for test to pass.

8.20 Step-Up Phase with Access Document with Early IssuerAuth

Table 8-33 NFC_RDR_NEG_STEPUP_AD_EARLY_ISSUER_AUTH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_EARLY_ISSUER_AUTH
PICS	Step-Up Phase AND Access Document processing – Validate time-based elements
Applicability	M for Reader that supports time concept and Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_EARLY_ISSUER_AUTH test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with early IssuerAuth. EXCHANGE command indicates transaction failure for test to pass.

8.21 Step-Up Phase with Access Document with Issuer Certificate Time Mismatch

Table 8-34 NFC_RDR_NEG_STEPUP_AD_ISSUER_CERTIFICATE_TIME_MISMATCH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_ISSUER_CERTIFICATE_TIME_MISMATCH
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_ISSUER_CERTIFICATE_TIME_MISMATCH test pre-conditions are identical to NFC_RDR_STEPUP_AD_ISSUER_CERT test pre-conditions in Table 8-19.

Repeat Table 8-17 with Access Document with Issuer Certificate validity time does not match “signed” date. EXCHANGE command indicates transaction failure for test to pass.

8.22 Step-Up Phase with Access Document with Validity Iteration

Table 8-35 NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION
PICS	Step-Up Phase AND

	Access Document verification – Validity Iteration
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Table 8-36 NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION test steps

Step#	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute NFC_RDR_STEPUP_AD_KEY_ID test steps (Table 8-17)		Verify the following: Tag 0x97 (Reader Status) value in EXCHANGE command matches the expected result. If all criteria are met, then CONTINUE else FAIL.
2	Repeat step 1 for each testable Access Document, in order.		Verify Criteria for all iterations is met. If all criteria are met, then PASS else FAIL.

Table 8-37 NFC_RDR_NEG_STEPUP_AD_VALIDITY_ITERATION test iterations

Iteration	Access Credential	Access Document	Expected Result
1	Access Credential 1	Minimal Document with Validity Iteration of 1	Access Granted (0x97 first byte is 0x01h)
2	Access Credential 2	Minimal Document with Validity Iteration of 9	Access Granted (0x97 first byte is 0x01h)
3	Access Credential 1	Minimal Document with Validity Iteration of 3	Access Granted (0x97 first byte is 0x01h)
4	Access Credential 3	Minimal Document with Validity Iteration of 1	Access Denied (0x97 first byte is 0x00h)

8.23 Step-Up Phase with Access Document with TimeVerificationRequired

Table 8-38 NFC_RDR_NEG_STEPUP_AD_TIME_VERIFICATION_REQUIRED test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_TIME_VERIFICATION_REQUIRED
PICS	Step-Up Phase AND Access Document verification – Validate time-based elements
Applicability	M for Reader that supports Step-Up Phase and does not support time concept

Interface	NFC
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NFC_RDR_NEG_STEPUP_AD_TIME_VERIFICATION_REQUIRED test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with Issuer Certificate with TimeVerificationRequired set and reader cannot determine time. EXCHANGE command indicates transaction failure for test to pass.

8.24 Step-Up Phase with Access Document with No Data Elements

Table 8-39 NFC_RDR_NEG_STEPUP_AD_NO_DATA_ELEMENTS test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_NO_DATA_ELEMENTS
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_NO_DATA_ELEMENTS test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with no data elements. EXCHANGE command indicates transaction failure for test to pass.

8.25 Step-Up Phase with Access Document with IssuerAuth docType Mismatch

Table 8-40 NFC_RDR_NEG_STEPUP_AD_ISSUER_DOCTYPE_MISMATCH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_ISSUER_DOCTYPE_MISMATCH
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_ISSUER_DOCTYPE_MISMATCH test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with IssuerAuth doctype does not match document docType. EXCHANGE command indicates transaction failure for test to pass.

8.26 Step-Up Phase with Access Document with docType Not Aliro-a

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_DOCTYPE_NOT_ALIROA

PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_DOCTYPE_NOT_ALIROA test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with doctype not “aliro-a”. EXCHANGE command indicates transaction failure for test to pass.

8.27 Step-Up Phase with Access Document with DeviceKeyInfo Mismatch

Table 8-41 NFC_RDR_NEG_STEPUP_AD_DEVICE_KEY_INFO_MISMATCH test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_DEVICE_KEY_INFO_MISMATCH
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_DEVICE_KEY_INFO_MISMATCH test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with deviceKeyInfo does not match Access Credential. EXCHANGE command indicates transaction failure for test to pass.

8.28 Step-Up Phase with Access Document with Invalid Access Data Element Version

Table 8-42 NFC_RDR_NEG_STEPUP_AD_INVALID_ACCESS_DATA_ELEMENT_VERSION test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_INVALID_ACCESS_DATA_ELEMENT_VERSION
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_INVALID_ACCESS_DATA_ELEMENT_VERSION test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with invalid access data element version. EXCHANGE command indicates transaction failure for test to pass.

8.29 Step-Up Phase with Access Document with No AccessRule for Intended Reader Action

Table 8-43 NFC_RDR_NEG_STEPUP_AD_NO_ACCESS_RULE_FOR_READER_ACTION test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_NO_ACCESS_RULE_FOR_READER_ACTION
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_NO_ACCESS_RULE_FOR_READER_ACTION test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with no AccessRule for intended reader action. EXCHANGE command indicates transaction failure for test to pass.

8.30 Step-Up Phase with Access Document with No Valid Schedule in AccessRule AllowScheduleIds

Table 8-44 NFC_RDR_NEG_STEPUP_AD_NO_VALID_SCHEDULE_ALLOW_SCHEDULEID test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_NO_VALID_SCHEDULE_ALLOW_SCHEDULEID
PICS	Step-Up Phase AND Access Data Element verification - Schedules
Applicability	M for Reader that supports Step-Up Phase and that supports schedules
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_NO_VALID_SCHEDULE_ALLOW_SCHEDULEID test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with no valid schedule in AccessRule AllowScheduleIds. EXCHANGE command indicates transaction failure for test to pass.

8.31 Step-Up Phase with Access Document with Valid Schedule in AccessRule DenyScheduleIds

Table 8-45 NFC_RDR_NEG_STEPUP_AD_VALID_SCHEDULE_DENY_SCHEDULEID test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_VALID_SCHEDULE_DENY_SCHEDULEID

PICS	Step-Up Phase AND Access Document processing - Schedules
Applicability	M for Reader that supports Step-Up Phase and that supports schedules
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_VALID_SCHEDULE_DENY_SCHEDULEID test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with valid schedule in AccessRule DenyScheduleIds. EXCHANGE command indicates transaction failure for test to pass.

8.32 Step-Up Phase with Access Document with Schedule in AccessRule and TimeVerifyRequired

Table 8-46 NFC_RDR_NEG_STEPUP_AD_SCHEDULE_TIME_VERIFY_REQUIRED test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_SCHEDULE_TIME_VERIFY_REQUIRED
PICS	Step-Up Phase AND NOT Access Document verification – Time-based elements
Applicability	M for Reader that supports Step-Up Phase and that does not support time-based elements
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_SCHEDULE_TIME_VERIFY_REQUIRED test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with schedule in AccessRule and TimeVerifyRequired. EXCHANGE command indicates transaction failure for test to pass.

8.33 Step-Up Phase with Access Document with Schedule in AccessRule with No Reader Support

Table 8-47 NFC_RDR_NEG_STEPUP_AD_SCHEDULE_IN_ACCESS_RULE_AND_READER test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_SCHEDULE_IN_ACCESS_RULE_AND_READER
PICS	Step-Up Phase AND NOT Access Document verification - Schedules
Applicability	M for Reader that supports Step-Up Phase and that does not support schedules
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_SCHEDULE_IN_ACCESS_RULE_AND_READER test pre-conditions are identical to NFC_RDR_STEPUP_AD_KEY_ID test pre-conditions in Table 8-16.

Repeat Table 8-17 with Access Document with schedule in AccessRule and Reader. EXCHANGE command indicates transaction failure for test to pass.

8.34 Step-Up Phase with Access Document with Unknown ReaderRule

Table 8-48 NFC_RDR_NEG_STEPUP_AD_UNKNOWN_READER_RULE test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_UNKNOWN_READER_RULE
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up Phase
Interface	NFC

Table 8-49 NFC_RDR_NEG_STEPUP_AD_UNKNOWN_READER_RULE test pre-conditions

Provision onto	Remarks
DUT (Reader)	IssuerKey_PubK, does not store a Reader Rule with ReaderRuleId 0xF118
TH (User Device)	reader_PubK, reader_group_identifier, Access Document

Repeat Table 8-17 with Access Document with unknown ReaderRule. EXCHANGE command indicates transaction failure for test to pass.

8.35 Step-Up Phase with Access Document with Unknown Critical AccessExtension

Table 8-50 NFC_RDR_NEG_STEPUP_AD_UNKNOWN_CRITICAL_ACCESS_EXTENSION test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_AD_UNKNOWN_CRITICAL_ACCESS_EXTENSION
PICS	Step-Up Phase AND Access Document processing
Applicability	M for Reader that supports Step-Up
Interface	NFC

NFC_RDR_NEG_STEPUP_AD_UNKNOWN_CRITICAL_ACCESS_EXTENSION test pre-conditions are identical to

NFC_RDR_STEPUP_AD_UNKNOWN_NON_ACCESS_EXTENSION test pre-conditions in Table 8-25.

Repeat Table 8-17 with Access Document with unknown critical AccessExtension. EXCHANGE command indicates transaction failure for test to pass.

8.36 Step-Up Phase with Revocation Document

Table 8-51 NFC_RDR_STEPUP_RD test identifiers

Parameter	Value
Test ID	NFC_RDR_STEPUP_RD
PICS	Step-Up Phase AND Revocation document storage and retrieval AND Revocation document processing
Applicability	M for Reader that supports Revocation Documents
Interface	NFC

Table 8-52 NFC_RDR_STEPUP_RD test pre-conditions

Provision onto	Remarks
TH (User Device)	reader_PubK, reader_group_identifier, Revocation Documents
DUT (Reader)	Access Credential long term public keys, IssuerKey_PubK

Table 8-53 NFC_RDR_STEPUP_RD test steps

Step#	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
4		Send SELECT command AID = A000000909ACCE5502, if Step-Up AID is required is indicated in signaling_bitmap in AUTH1 response	If all criteria are met, then CONTINUE else FAIL.
5	Send SELECT response		
6		Request Revocation Document using DeviceRequest inside ENVELOPE command.	Verify the following: Revocation Document is requested in ENVELOPE command response. If all criteria are met, then CONTINUE else FAIL.
7	Send appropriate Revocation Document in DeviceResponse inside ENVELOPE command response		
8	One or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		

Step#	TH (User Device)	DUT (Reader)	Verification at TH
9	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then CONTINUE else FAIL.
10	Wait for at least 3 seconds.		
	Execute NFC_RDR_STEPUP_AD_KEY_ID test steps (Table 8-17) with Access Credential 1		Verify the following: Tag 0x97 matches expected result for Access Credential 1. If all criteria are met, then CONTINUE else FAIL.
11	Wait for at least 3 seconds.		
12	Execute NFC_RDR_STEPUP_AD_KEY_ID test steps (Table 8-17) with Access Credential 2		Verify the following: Tag 0x97 matches expected result for Access Credential 2. If all criteria are met, then CONTINUE else FAIL.
13	Steps 1-12 are repeated for each testable Revocation Document, in order		Verify the following: Criteria for all iterations is met. If all criteria are met, then PASS else FAIL.

Table 8-54 NFC_RDR_STEPUP_RD test iterations

Iteration	Revocation Document	Access Credential 1 Result	Access Credential 2 Result
1	Overwrite containing Access Credential 1 Public Key	Access Rejected (0x97 first byte is 0x00h)	Access Accepted (0x97 first byte is 0x01h)
2	Update adding Access Credential 2 Public Key	Access Rejected	Access Rejected
3	Update removing Access Credential 1 Public Key	Access Accepted	Access Rejected
4	Update adding Access Credential 1 Public Key and removing Access Credential 1 Public Key and Access Credential 2 Public Key	Access Accepted	Access Accepted
5	Overwrite containing Access Credential 2 Public Key	Access Accepted	Access Rejected
6	Overwrite empty	Access Accepted	Access Accepted

8.37 Step-Up Phase with Revocation Document with Invalid Revocation Document Version

Table 8-55 NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION
PICS	Step-Up Phase AND

	Revocation document storage and retrieval AND Revocation document processing
Applicability	M for Reader that supports Revocation Documents
Interface	NFC

NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION test pre-conditions are identical to NFC_RDR_STEPUP_RD test pre-conditions in Table 8-52
NFC_RDR_STEPUP_RD test pre-conditions

Table 8-56 NFC_RDR_NEG_STEPUP_RD_INVALID_ELEMENT_VERSION test steps

Step#	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2).		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
4		Send SELECT command AID = A000000909ACCE5502, if Step-Up AID is required is indicated in signaling_bitmap in AUTH1 response.	If all criteria are met, then CONTINUE else FAIL.
5	Send SELECT response.		
6		Request Revocation Document using DeviceRequest inside ENVELOPE command.	Verify the following: Revocation Document is requested in ENVELOPE command response. If all criteria are met, then CONTINUE else FAIL.
7	Send ENVELOPE command response.		
8	One or more GET RESPONSE command and GET RESPONSE command response can be exchanged.		
9	Execute EXCHANGE indicating transaction failure routine (Table 6-5).		If all criteria are met, then PASS else FAIL.

8.38 SELECT Response with No Common Expedited Protocol Version

Table 8-57 NFC_RDR_NEG_SEL_RSP_NO_COMMON_EXPEDITED_PROTOCOL_VERSION test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_SEL_RSP_NO_COMMON_EXPEDITED_PROTOCOL_VERSION
PICS	Expedited-Standard Phase
Applicability	M for Reader

Interface	NFC
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NFC_RDR_NEG_SEL_RSP_NO_COMMON_EXPEDITED_PROTOCOL_VERSION test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-58 NFC_RDR_NEG_SEL_RSP_NO_COMMON_EXPEDITED_PROTOCOL_VERSION test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send SELECT command.	Verify the following: - AID = A000000909ACCE5501 - Format of command matches the specification. If all criteria are met, then CONTINUE else FAIL.
2	Send SELECT Response. Set the expedited_phase_supported_protocol_versions equal to 0x0A00		
3		Send CONTROL FLOW command.	Verify the following: - CONTROL FLOW command data field length does not exceed 255 bytes. - S2_parameter in command data field is equal to 0x27. If all criteria are met, then CONTINUE else FAIL.
4	send CONTROL FLOW response.		If all criteria are met, then PASS else FAIL.

8.39 AUTH0 with Extra Unknown TLV

Table 8-59 NFC_RDR_NEG_AUTH0_EXTRA_TAG test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_AUTH0_EXTRA_TAG
PICS	Expedited-Standard Phase
Applicability	M for Reader
Interface	NFC

NFC_RDR_NEG_AUTH0_EXTRA_TAG test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-60 NFC_RDR_NEG_AUTH0_EXTRA_TAG test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). Extra tag can be randomly injected at any location in the AUTH0 response command payload.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3).		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.40 AUTH0 with Wrong Value

Table 8-61 NFC_RDR_NEG_AUTH0_WRONG_VALUE test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_AUTH0_WRONG_VALUE
PICS	Expedited-Standard Phase
Applicability	M for Reader
Interface	NFC

NFC_RDR_NEG_AUTH0_WRONG_VALUE test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-62 NFC_RDR_NEG_AUTH0_WRONG_VALUE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). wrong value/length of tag in the AUTH0 response command payload.		If all criteria are met, then CONTINUE else FAIL.
3	Execute CONTROL FLOW indicating transaction failure routine (Table 6-6).		If all criteria are met, then PASS else FAIL.

8.41 AUTH1 with Wrong User Device Signature

Table 8-63 NFC_RDR_NEG_AUTH1_WRONG_UD_SIGNATURE test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_AUTH1_WRONG_UD_SIGNATURE
PICS	Expedited-Standard Phase
Applicability	M for Reader

Interface	NFC
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NFC_RDR_NEG_AUTH1_WRONG_UD_SIGNATURE test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-64 NFC_RDR_NEG_AUTH1_WRONG_UD_SIGNATURE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3). Send wrong User Device signature in AUTH1 response.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction failure routine (Table 6-5).		If all criteria are met, then PASS else FAIL.

8.42 AUTH1 with Extra Tag

Table 8-65 NFC_RDR_NEG_AUTH1_EXTRA_TAG test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_AUTH1_EXTRA_TAG
PICS	Expedited-Standard Phase
Applicability	M for Reader
Interface	NFC

NFC_RDR_NEG_AUTH1_EXTRA_TAG test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-66 NFC_RDR_NEG_AUTH1_EXTRA_TAG test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3). Send extra unknown tag TLV in AUTH1 response.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 6-4).		If all criteria are met, then PASS else FAIL.

8.43 AUTH1 with Wrong Values

Table 8-67 NFC_RDR_NEG_AUTH1_WRONG_VALUES test identifiers

Parameter	Value
Test ID	NFC_RDR_NEG_AUTH1_WRONG_VALUES
PICS	Expedited-Standard Phase
Applicability	M for Reader
Interface	NFC

NFC_RDR_NEG_AUTH1_WRONG_VALUES test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-68 NFC_RDR_NEG_AUTH1_WRONG_VALUES test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3). Send wrong value/length for tag in AUTH1 response.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction failure routine (Table 6-5).		If all criteria are met, then PASS else FAIL.

8.44 EXCHANGE with Reader Descriptor Tag

Table 8-69 NFC_RDR_EXCHANGE_RDR_DESCRIPTOR_TAG test identifiers

Parameter	Value
Test ID	NFC_RDR_EXCHANGE_RDR_DESCRIPTOR_TAG
PICS	Reader Descriptor tag
Applicability	M for Reader that supports sending Reader Information to the User Device
Interface	NFC

NFC_RDR_EXCHANGE_RDR_DESCRIPTOR_TAG test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-70 NFC_RDR_EXCHANGE_RDR_DESCRIPTOR_TAG test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). Set AID = A000000909ACCE5501.		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
3	Execute AUTH1 routine (Table 6-3). Send extra unknown tag TLV in AUTH1 response.		If all criteria are met, then CONTINUE else FAIL.
4	Execute EXCHANGE indicating transaction success routine (Table 6-4).		Verify the following in addition to EXCHANGE routine: - order of TLVs in EXCHANGE command matches specification. - All mandatory TLVs in EXCHANGE command present 0xAE with sub tag 0xB5 is present - Format of 0xB5 matches the technical specification 0xAE length is less than 250 bytes If all criteria are met, then PASS else FAIL.

8.45 Control Flow with Reader Descriptor Tag

Parameter	Value
Test ID	NFC_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG
PICS	Reader Descriptor tag
Applicability	M for Reader that supports sending Reader Information to the User Device
Interface	NFC

NFC_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-71 NFC_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute NFC_RDR_NEG_AUTH0_WRONG_VALUE test steps (Table 8-62).		Verify the following in addition to NFC_RDR_NEG_AUTH0_WRONG_VALUE test steps: - All mandatory TLVs in Control Flow command are present. 0x63 is present. - Format of 0x63 matches the technical specification. - Control Flow command data field length is less than 255 bytes. If all criteria are met, then PASS else FAIL.

8.46 BLE+UWB Flow with Reader Descriptor Tag

Table 8-72 BLEUWB_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG
PICS	BLE + UWB Flow AND Reader Descriptor tag
Applicability	M for Reader that supports sending Reader Information to the User Device
Interface	BLE

BLEUWB_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG test pre-conditions are identical to NFC_RDR_STANDARD_NO_CERT test pre-conditions in Table 8-2.

Table 8-73 BLEUWB_RDR_CONTROL_FLOW_RDR_DESCRIPTOR_TAG test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection		
3	Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501		
4	Execute AUTH0 routine (Table 6-2). Send wrong value/length of tag in AUTH0 response.		If all criteria are met, then CONTINUE else FAIL.
5		send Event Message ID carrying General Error Attribute ID and Reader Information Attribute ID	Verify the following: 1. General Error Attribute ID and Reader Descriptor Attribute ID are present 2. Format of Attribute IDs matches the technical specification If all criteria are met, then PASS else FAIL.
6	BLE teardown		

8.47 EXCHANGE with Mailbox Command

This test assumes Reader Under Test can be made to send EXCHANGE with Mailbox commands.

Table 8-74 NFC_RDR_EXCHANGE_MAILBOX test identifiers

Parameter	Value
Test ID	NFC_RDR_EXCHANGE_MAILBOX
PICS	Mailbox
Applicability	M for Reader that supports Mailbox commands
Interface	NFC

Table 8-75 NFC_RDR_EXCHANGE_MAILBOX test pre-conditions

Provision onto	Remarks
TH (User Device)	Reader_PubK, reader_group_identifier, non-zero mailbox populated with existing data
DUT (Reader)	Access Credential long term public key

Table 8-76 NFC_RDR_EXCHANGE_MAILBOX test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute SELECT routine (Table 6-1). AID = A000000909ACCE5501		If all criteria are met, then CONTINUE else FAIL.
2	Execute AUTH0 routine (Table 6-2). Set command_parameters = 0h.		If all criteria are met, then CONTINUE else FAIL.
3	Execute AUTH1 routine (Table 6-3). Reader_Cert is not presented AUTH1 command.		If all criteria are met, then CONTINUE else FAIL.
4		Send EXCHANGE command multiple times with multiple Mailbox commands to or from mailbox	Verify the following: Request format matches technical specification. Requests contain one or more Mailbox commands. If all criteria are met, then CONTINUE else FAIL.
5	Send EXCHANGE response		
6	Verify mailbox contents		Verify the following: Mailbox content format matches the technical specification. If all criteria are met, then CONTINUE else FAIL.

8.48 BLE+UWB Flow with Expedited Standard Phase

Table 8-77 BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test identifiers

Parameter	Value
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Test ID	BLEUWB_RDR_EXPEDITED_STANDARD_PHASE
PICS	BLE + UWB Flow AND UWB ranging AND Dynamic Advertisement Tag AND Unsolicited reader status reporting AND Expedited-Standard Phase AND UWB Time Synchronization
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

Table 8-78 BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions

Provision onto	Remarks
DUT (Reader)	Access Credential long term public key, GRK
TH (User Device)	reader_PubK, reader_group_identifier, GRK
NOTE 1: The TH (User Device) and the DUT (Reader) are in very close proximity (e.g., 1 m and line-of-sight).	
NOTE 2: Reader is in secured state as a pre-condition.	

Table 8-79 BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 6-7).		If all criteria are met, then CONTINUE else FAIL.
2	Send Time Sync Message ID		
3	Execute BLE+UWB Ranging Session setup routine (Table 6-8).		If all criteria are met, then CONTINUE else FAIL.
4	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
5		Send Reader Status Changed Message ID carrying State Attribute ID	Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

8.49 BLE+UWB Flow with Expedited Fast Phase

Table 8-80 BLEUWB_RDR_EXPEDITED_FAST_PHASE test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_EXPEDITED_FAST_PHASE
PICS	BLE + UWB Flow AND UWB ranging AND Dynamic Advertisement Tag AND Unsolicited reader status reporting AND Expedited-fast
Applicability	M for Reader that supports BLE + UWB Flow and Expedited-Fast Phase
Interface	BLE

BLEUWB_RDR_EXPEDITED_FAST_PHASE test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-81 BLEUWB_RDR_EXPEDITED_FAST_PHASE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 6-7).		If all criteria are met, then CONTINUE else FAIL.
2	BLE teardown.		
3		Send Bluetooth LE advertisement	
4	Establish L2CAP connection		
5	Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501		
6		Send AUTH0 command with command_parameters = 1h	
7	Send AUTH0 response		
8		Send EXCHANGE command with Tag 0x98	Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.
9	Send EXCHANGE Response		
10		Send Reader Status Access Protocol Completed Message ID	Verify the following: Ensure reader status is secured.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
		carrying Reader Information Attribute ID	Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
11	Send Time Sync Message ID		
12	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
13	Execute BLE+UWB ranging session setup routine (Table 6-8).		If all criteria are met, then CONTINUE else FAIL.
14	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
15		Reader Status Changed Message ID carrying State Attribute ID is sent	Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

8.50 BLE+UWB Flow with Step-Up Phase

Table 8-82 BLEUWB_RDR_STEPUP_PHASE test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_STEPUP_PHASE
PICS	BLE + UWB Flow AND UWB ranging AND Dynamic Advertisement Tag AND Unsolicited reader status reporting AND Step-Up Phase
Applicability	M for Reader that supports BLE + UWB Flow and support Step-Up Phase
Interface	BLE

Table 8-83 BLEUWB_RDR_STEPUP_PHASE test pre-conditions

Provision onto	Remarks
DUT (Reader)	Access Credential long term public key, GRK, IssuerKey_PubK
TH (User Device)	reader_PubK, reader_group_identifier, GRK, Access Document
NOTE 1: The TH (User Device) and the DUT (Reader) are in very close proximity (e.g., 1 m and line-of-sight).	
NOTE 2: Reader is in secured state as a pre-condition.	

Table 8-84 BLEUWB_RDR_STEPUP_PHASE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is set to 1. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection.		
3	Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501		
4	Execute AUTH0 routine. command_parameters = 0h authentication_policy = 01h (User Device)		If all criteria are met, then CONTINUE else FAIL.
5	Execute AUTH1 routine.		If all criteria are met, then CONTINUE else FAIL.
6		Send EXCHANGE command	Verify the following: Tag 0x98 is present. If all criteria are met, then CONTINUE else FAIL.
7	Send EXCHANGE response		
8		Request Access Document using DeviceRequest inside ENVELOPE command	
9	Send Access Document in DeviceResponse inside ENVELOPE command response		

Steps	TH (User Device)	DUT (Reader)	Verification at TH
10	One or more GET RESPONSE command/response can be exchanged		
11		Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID	Verify the following: Ensure reader status is secured. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
12	Send Time Sync Message ID		
13	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
14	Execute BLE+UWB ranging session setup routine (Table 6-8).		If all criteria are met, then CONTINUE else FAIL.
15	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
16		Reader Status Changed Message ID carrying State Attribute ID is sent	Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

8.51 BLE+UWB Flow with UWB Ranging Suspend

Table 8-85 BLEUWB_RDR_RANGING_SUSPEND test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_RANGING_SUSPEND
PICS	BLE + UWB Flow AND UWB ranging suspend
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_RANGING_SUSPEND test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78 with the exception: The TH (User Device) and the DUT (Reader) are in close proximity (e.g., 5 m and line-of-sight).

Table 8-86 BLEUWB_RDR_RANGING_SUSPEND test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 6-7).		If all criteria are met, then CONTINUE else FAIL.
2	Send Time Sync Message ID		
3	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
4	Execute BLE+UWB ranging session setup routine (Table 6-8).		If all criteria are met, then CONTINUE else FAIL.
5	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
6	Send Ranging Session Suspend Request with correct UWB Session Identifier		
7		Send Ranging Session Suspend Response	Verify the following 1. this message is sent. 2. format of Ranging Session Suspend Response matches technical specification. If all criteria are met, then PASS else FAIL. The status can either value 0 or 1.

8.52 BLE+UWB Flow with UWB Ranging Resume

Parameter	Value
Test ID	BLEUWB_RDR_RANGING_RESUME
PICS	BLE + UWB Flow AND UWB ranging resume
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_RANGING_RESUME test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78 with the exception: The TH (User Device) and the DUT (Reader) are in close proximity (e.g., 5 m and line-of-sight).

Table 8-87 BLEUWB_RDR_RANGING_RESUME test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 6-7).		If all criteria are met, then CONTINUE else FAIL.
2	Send Time Sync Message ID		
3	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
4	Execute BLE+UWB ranging session setup routine (Table 6-8).		If all criteria are met, then CONTINUE else FAIL.
5	Allow N (e.g., 3 seconds) for UWB ranging to occur		Verify the following: UWB packets are exchanged over UWB transport. If all criteria are met, then CONTINUE else FAIL.
6	Send Ranging Message ID carrying Ranging Session Suspended Attribute ID		
7	1 second after previous step, send Ranging Message ID carrying Initiate Ranging Session Resume Attribute ID		
8		send Ranging Session Resume Request Message	NOTE: For this test, the User Device and Reader are next close (e.g., 5 m and line-of-sight) to each other.
9	Send Ranging Session Resume Response Message		Verify the following: 1. Format of the message matches technical specification. 2. UWB ranging is resumed. 3. UWB packets are exchanged over UWB transport in short time (e.g., up to 3 seconds) after sending Ranging Session Resume Response Message. If all criteria are met, then PASS else FAIL.

8.53 BLE+UWB Flow with Failed L2CAP

Table 8-88 BLEUWB_RDR_NEG_FAILED_L2CAP test identifiers

Parameter	Value
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Test ID	BLEUWB_RDR_NEG_FAILED_L2CAP
PICS	BLE + UWB Flow
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_NEG_FAILED_L2CAP test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-89 BLEUWB_RDR_NEG_FAILED_L2CAP test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	send wrong Selected Aliro Ble UWB Protocol Version	Establish L2CAP connection fails	Verify the following: L2CAP establishment fails. If all criteria are met, then PASS else FAIL.

8.54 BLE+UWB Flow with wrong SPSM

Table 8-90 BLEUWB_RDR_NEG_FAILED_SPSM_L2CAP test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_NEG_FAILED_SPSM_L2CAP
PICS	BLE + UWB Flow
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_NEG_FAILED_SPSM_L2CAP test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-91 BLEUWB_RDR_NEG_FAILED_SPSM_L2CAP test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	send wrong SPSM value	Establish L2CAP connection fails	Verify the following: L2CAP establishment fails.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
			If all criteria are met, then PASS else FAIL.

8.55 BLE+UWB Flow with timeout before AUTH0

Table 8-92 BLEUWB_RDR_NEG_TIMEOUT_BEFORE_AUTH0 test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_NEG_TIMEOUT_BEFORE_AUTH0
PICS	BLE + UWB Flow
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_NEG_TIMEOUT_BEFORE_AUTH0 test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-93 BLEUWB_RDR_NEG_TIMEOUT_BEFORE_AUTH0 test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection		
3	Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501		
4		Send AUTH0 command	
5	Wait for at least 3 seconds before sending AUTH0 command response		
6		Send Event Message ID carrying General Error Attribute ID	Verify the following: Format of Event Message ID carrying General Error Attribute ID matches the technical specification. If all criteria are met, then CONTINUE else FAIL.
7		BLE teardown	Verify the following: BLE teardown is initiated by the DUT (Reader).

Steps	TH (User Device)	DUT (Reader)	Verification at TH
			If all criteria are met, then PASS else FAIL.

8.56 BLE+UWB Flow with Timeout Extension

Table 8-94 BLEUWB_RDR_TIMEOUT_EXTENSION test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_TIMEOUT_EXTENSION
PICS	BLE + UWB Flow
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_TIMEOUT_EXTENSION test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-95 BLEUWB_RDR_TIMEOUT_EXTENSION test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE + UWB Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection		
3	Send Initiate Access Protocol Message ID carrying AID = A000000909ACCE5501		
4		Send AUTH0 command	
5	Send Event carrying Busy Attribute ID at 1 s after receiving AUTH0 command		
6	Send AUTH0 command response after 1 s after sending Event Busy Attribute ID		
7		Send AUTH1 command	
8	Send AUTH1 response		
9		Send EXCHANGE with Tag 0x98	Verify the following: Tag 0x98 is present.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
			If all criteria are met, then CONTINUE else FAIL.
10	Send EXCHANGE response		
11		Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID	Verify the following: Format of message matches technical specification. If all criteria are met, then PASS else FAIL.

8.57 BLE+UWB Flow with M2 Message Mismatch Parameter

Table 8-96 BLEUWB_RDR_NEG_M2_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_NEG_M2_MISMATCH_PARAMETER
PICS	BLE + UWB Flow
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_NEG_M2_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-97 BLEUWB_RDR_NEG_M2_MISMATCH_PARAMETER test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 6-7).		If all criteria are met, then CONTINUE else FAIL.
2	Send Time Sync Message ID		
4	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
5		Send Ranging Session Setup M1 Message ID	
6	Send Ranging Session Setup M2 Message ID without UWB Config ID		
7		Send Event with General Error Attribute ID indicating Wrong Parameters	Verify the following: Format of Event Message ID. General Error Attribute ID indicates Wrong Parameters.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
			If all criteria are met, then PASS else FAIL.
8	BLE teardown		

8.58 BLE+UWB Flow with M4 Message Mismatch Parameter

Table 8-98 BLEUWB_RDR_NEG_M4_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_NEG_M4_MISMATCH_PARAMETER
PICS	BLE + UWB Flow
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_NEG_M4_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-99 BLEUWB_RDR_NEG_M4_MISMATCH_PARAMETER test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE+UWB Aliro Access Protocol routine (Table 6-7).		If all criteria are met, then CONTINUE else FAIL.
2	Send Time Sync Message ID		
3	Send Ranging Message ID carrying Initiate Ranging Session Attribute ID		
4		Send Ranging Session Setup M1 Message ID	
5	Send Ranging Session Setup M2 Message		
6		Send Ranging Session Setup M3 Message	
7	Send Ranging Session Setup M4 Message ID without UWB Time0		
15		Send Ranging Message ID carrying Secure Ranging Over UWB Radio Failed Attribute ID or General Error	Verify the following: Format of this message matches the specification. If all criteria are met, then PASS else FAIL.

8.59 BLE+UWB Flow with Suspend Request Mismatch Parameter

Table 8-100 BLEUWB_RDR_NEG_SUSPEND_MISMATCH_PARAMETER test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_NEG_SUSPEND_MISMATCH_PARAMETER
PICS	BLE + UWB Flow AND UWB Ranging Suspend
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

BLEUWB_RDR_NEG_SUSPEND_MISMATCH_PARAMETER test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-101 BLEUWB_RDR_NEG_SUSPEND_MISMATCH_PARAMETER test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLEUWB_RDR_RANGING_SUSPEND test steps (Table 8-86). Send Ranging Session Suspend Request without UWB Session Identifier	Send Event Message ID carrying General Error Attribute ID indicating Wrong Parameters	Verify the following: Format of Event Message ID. General Error Attribute ID indicates Wrong Parameters. If all criteria are met, then PASS else FAIL.

8.60 BLE+UWB Flow BLE Advertisement Format

Table 8-102 BLEUWB_RDR_ADVERTISEMENT_FORMAT test identifiers

Parameter	Value
Test ID	BLEUWB_RDR_ADVERTISEMENT_FORMAT
PICS	BLE + UWB Flow AND Dynamic advertisement tag
Applicability	M for Reader that supports BLE + UWB Flow
Interface	BLE

Table 8-103 BLEUWB_RDR_ADVERTISEMENT_FORMAT test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: advertisement data matches technical specification. If all criteria are met, then PASS else FAIL.

8.61 BLE-only Flow – RKE Unsecure

Table 8-104 BLERKE_RDR_UNSECURE test identifiers

Parameter	Value
Test ID	BLERKE_RDR_UNSECURE
PICS	BLE-Only Flow AND Unsolicited reader status reporting
Applicability	M for Reader that supports BLE-Only Flow
Interface	BLE

BLERKE_RDR_UNSECURE test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78. Reader is in secured state before running this test.

Table 8-105 BLERKE_RDR_UNSECURE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE-Only Aliro Access Protocol routine (Table 6-9).		If all criteria are met, then CONTINUE else FAIL.
2	Send RKE Request Message ID with action=UNSECURE		
3		Send Reader Status Changed Message ID carrying State Attribute ID (Unsecured)	Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.

8.62 BLE-only Flow – RKE Secure

Table 8-106 BLERKE_RDR_SECURE test identifiers

Parameter	Value
Test ID	BLERKE_RDR_SECURE
PICS	BLE-Only Flow AND Unsolicited reader status reporting
Applicability	M for Reader that supports BLE-Only Flow
Interface	BLE

BLERKE_RDR_SECURE test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78. Reader is in unsecured state before running this test.

Table 8-107 BLERKE_RDR_SECURE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLE-Only Aliro Access Protocol routine (Table 6-9).		If all criteria are met, then CONTINUE else FAIL.
11	Send RKE Request Message ID with action=SECURE		
12		Send Reader Status Changed Message ID carrying State Attribute ID (Secured)	Verify the following: State Attribute ID has second byte [B7:B0] set to 0x00 or 0x02 or 0x80 or 0x82. If all criteria are met, then PASS else FAIL.

8.63 BLE-Only Flow with Disallowed Expedited Fast Phase

Table 8-108 BLERKE_RDR_NEG_FAST test identifiers

Parameter	Value
Test ID	BLERKE_RDR_NEG_FAST
PICS	BLE-Only Flow
Applicability	M for Reader that supports BLE-Only Flow
Interface	BLE

BLERKE_RDR_NEG_FAST test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-109 BLERKE_RDR_NEG_FAST test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1	Execute BLERKE_RDR_UNSECURE test steps (Table 8-105) without changing the test pre-conditions for BLERKE_RDR_NEG_FAST test.		If all criteria are met, then CONTINUE else FAIL.
2	Secure the Reader and BLE teardown		
3	Repeat steps 1 and 2, ten times.		Verify the following: Each iteration passes. If all criteria are met, then PASS else FAIL

8.64 BLE-Only Flow with Failed L2CAP

Table 8-110 BLERKE_RDR_NEG_FAILED_L2CAP test identifiers

Parameter	Value
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Test ID	BLERKE_RDR_NEG_FAILED_L2CAP
PICS	BLE-Only Flow
Applicability	M for Reader that supports BLE-Only Flow
Interface	BLE

BLERKE_RDR_NEG_FAILED_L2CAP test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-111 BLERKE_RDR_NEG_FAILED_L2CAP test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE-Only Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	send wrong Selected Aliro Ble UWB Protocol Version	Establish L2CAP connection fails	Verify the following: L2CAP setup fails. If all criteria are met, then PASS else FAIL.

8.65 BLE-Only Flow with wrong SPSM

Table 8-112 BLERKE_RDR_NEG_FAILED_SPSM_L2CAP test identifiers

Parameter	Value
Test ID	BLERKE_RDR_NEG_FAILED_SPSM_L2CAP
PICS	BLE-Only Flow
Applicability	M for Reader that supports BLE-Only Flow
Interface	BLE

BLERKE_RDR_NEG_FAILED_SPSM_L2CAP test pre-conditions are identical to BLEUWB_RDR_EXPEDITED_STANDARD_PHASE test pre-conditions in Table 8-78.

Table 8-113 BLERKE_RDR_NEG_FAILED_SPSM_L2CAP test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE-Only Aliro Flow Supported Bit is not set to 1. If all criteria are met, then CONTINUE else FAIL.
2	send wrong SPSM value	Establish L2CAP connection fails	Verify the following: L2CAP setup fails. If all criteria are met, then PASS else FAIL.

8.66 BLE-Only Flow with Step-Up Phase

Table 8-114 BLERKE_RDR_STEPUP_PHASE test identifiers

Parameter	Value
Test ID	BLERKE_RDR_STEPUP_PHASE
PICS	BLE-Only Flow
Applicability	M for Reader that supports BLE-Only Flow AND supports Step-Up Phase
Interface	BLE

BLERKE_RDR_STEPUP_PHASE test pre-conditions are identical to BLEUWB_RDR_STEPUP_PHASE test pre-conditions in Table 8-83.

Table 8-115 BLERKE_RDR_STEPUP_PHASE test steps

Steps	TH (User Device)	DUT (Reader)	Verification at TH
1		Send Bluetooth LE advertisement	Verify the following: BLE-Only Aliro Flow Supported Bit is set to 1. Advertisement format matches the technical specification. If all criteria are met, then CONTINUE else FAIL.
2	Establish L2CAP connection		
3	Send Initiate Access Protocol RKE Message ID carrying AID = A000000909ACCE5501		
4		Send AUTH0 command	Verify the following: command_parameters = 0h. authentication_policy != 02h authentication_policy = 01h or 03h If all criteria are met, then CONTINUE else FAIL.
5	Send AUTH0 response		
6		Send AUTH1 command	
7	Send AUTH1 response		
8		[Optional] Send EXCHANGE command	
9	Send EXCHANGE response		Sent, if an EXCHANGE command is received.

Steps	TH (User Device)	DUT (Reader)	Verification at TH
10		Request Access Document using DeviceRequest inside ENVELOPE command	
11	Send Access Document in DeviceResponse inside ENVELOPE command response		
12	One or more GET RESPONSE command/response can be exchanged		
13		Send Reader Status Access Protocol Completed Message ID carrying Reader Information Attribute ID	Verify the following: Ensure reader status is secured. Format of message matches technical specification. If all criteria are met, then CONTINUE else FAIL.
14	Send RKE Request Message ID with action=UNSECURE		
15		Reader Status Changed Message ID carrying State Attribute ID is sent	Verify the following: State Attribute ID has second byte [B7:B0] set to 0x01 or 0x02 or 0x81 or 0x82. If all criteria are met, then PASS else FAIL.